

# The classical Rankine-Hugoniot jump conditions, an important cornerstone of modern shock wave physics: ideal assumptions vs. reality

Peter O.K. Krehl<sup>a,b</sup>

79331 Nimbург, Baden-Württemberg, Germany

Received 29 January 2014 / Received in final form 11 February 2015

Published online 2 April 2015

© EDP Sciences, Springer-Verlag 2015

**Abstract.** The purpose of this paper is to discuss from a historical point of view the classical Rankine-Hugoniot (RH) relations in more detail than usually done in standard textbooks. Particularly focusing on the last seventy years, this paper (i) reviews their validity and limitations as interpreted by numerous users; (ii) summarizes their enormous extension also to other branches of science and engineering; and (iii) discusses the nontrivial problem of error estimation. Originally, the RH relations were derived for a plane-parallel steadily propagating aerial shock with a step wave profile; i.e., a wave with zero rise-time and constant thermodynamic as well as kinematic parameter values behind the shock front. But real shock waves are in most cases three-dimensional, have finite rise-times, and in almost all cases are unsteady waves. These real properties must produce a systematic error when applying the RH relations and the cardinal question arises how large this error will be in comparison with a random error caused by the applied high-speed diagnostics. However, numerical procedures of studying systematic errors of unsteady wave propagation are difficult to carry out because of various reasons and still pending.

## 1 Introduction

The famous *Rankine-Hugoniot (RH) relations* [1,2], an ingenious and simple concept based upon the conservation laws of mass, momentum and energy<sup>1</sup>, are widely used in

---

<sup>a</sup> Dr. Krehl is a retired staff scientist from the Fraunhofer-Institute for High-Speed Dynamics, Ernst-Mach-Institut (FhG-EMI) at Freiburg/Breisgau, Germany.

<sup>b</sup> e-mail: [peter.o.k.krehl@t-online.de](mailto:peter.o.k.krehl@t-online.de)

<sup>1</sup> The conservation laws were first recognized in the early stages of development of classical mechanics. Their generality and simplicity lie in the fact that they apply and may be exploited even in situations for which one is ignorant of the details of internal mechanisms – in the example of the RH relations the inner structure of the shock transition. The conservation laws, an important class of partial differential equations, are now believed to be universally valid also for some other physical quantities. For example, for collisionless hydrodynamic shocks additional jump conditions for electromagnetic fields, electrons, ions and the bulk

shock physics, because they quickly provide data of shock parameter values for many practical applications rather than obtaining more precise data by elaborately solving hyperbolic differential equations. The RH relations are one of the cornerstones of modern shock physics and certainly also contributed considerably to the application and analysis of shock waves in other branches of science and engineering (S&E) – meanwhile ranging from microscopic to cosmic dimensions, thus turning shock wave physics into an interdisciplinary field in its own right [3]. The RH relations are widely applied in scientific research such as in fluid and gas dynamics, materials science, relativity theory, quantum mechanics, aerodynamics, ocean dynamics, meteorology, and astrophysics. In computational fluid dynamics, the third RH relation, known as the *RH condition*, is used in the discretization process for numerically solving hyperbolic systems of conservation laws [4].

The classical RH relations – in a way the “Holy Grail” of shock physics – are a set of three equations relating the state variables of the shocked medium to the ones of the undisturbed medium. Originally, they were derived for a plane-parallel *steady shock*<sup>2</sup>, thereby neglecting any viscosity and heat conduction effects within and behind the shock front. Such a wave has a step profile; i.e., is a wave with zero rise-time and constant thermodynamic and kinematic variables behind the shock front. However, real shock waves, whether man-made or naturally occurring on Earth or in space, are generally three-dimensional, have finite rise-times, are subjected to energy dissipation (usually caused by viscosity), and in almost all cases are unsteady waves. Typically, they change their pressure-time profile during propagation and in most cases are far from being step functions in time. Nevertheless, in spite of these discrepancies with the RH concept, some shock researchers, encouraged by their observed good agreement between calculated and measured shock parameters, considered the RH relations implicitly as “exact” and didn’t question their general validity also to shock waves with finite rise-times and unsteady propagation. This (incorrect) view, particularly spread among gas dynamicists, might have arisen from early shock tube measurements at Princeton University [5] which showed a total error below 1% between measured data and calculated data using the RH relations – an exception to most practical examples because the shock-time profile of a shock wave propagating in a shock tube closely matches a step function.

The RH relations have several unique features:

- (i) All three equations do not require information on the equation of state (EOS); therefore, they are applicable to any state of aggregation, even to multiphase media and media undergoing phase transitions.
- (ii) They relate kinematic parameters ( $u_0$ ,  $u_1$ ,  $U_s$ ) to thermodynamic parameters ( $\rho_0$ ,  $\rho_1$ ,  $P_0$ ,  $P_1$ ,  $e_0$ ,  $e_1$ ) across the shock transition<sup>3</sup>. In most cases the parameter values for the initial state are well-known so that only five unknown shock variables remain, independently of the chosen coordinates.
- (iii) By experimentally determining two variables of the shocked state, the other three variables can be calculated using the three RH equations. Since kinematic properties can be measured with a higher precision than thermodynamic ones, the RH relations are favorably used to determine EOS in shock-compressed materials, known as *Hugoniot*s. In the special case of an ideal gas, only a single parameter value, the shock front velocity or Mach number, is necessary to determine all the other shock parameter values.

---

plasma have been derived. Early studies of the Euler equations in gas dynamics initiated the *Theory of Hyperbolic Conservation Laws*.

<sup>2</sup> See also Section 5.1, #12.

<sup>3</sup> For symbols and subscripts see Section 2.2.1.

- (iv) The first two RH equations require no assumptions about the dynamic equilibrium and apply throughout the shock transition layer; i.e., a shock jump can be treated without knowing any details of the shock structure – a supposition useful for most practical applications, but only an approximation.
- (v) The third RH equation – a thermodynamic condition among fluid dynamicists and mathematicians known as the *Rankine-Hugoniot condition* – considers conservation of energy which allows one to also make some estimations on temperature and entropy jumps across the shock front. Since the third equation which is independent of the chosen coordinate system contains no velocity terms, only thermodynamic quantities, it is widely used in S&E.
- (vi) The RH relations hold not only across shock waves but also across *contact discontinuities*<sup>4</sup> such as during particular cases of obliquely interacting shock wave known as *Mach reflection* [6], in traffic flow modeling [7] and even in supernova remnants separating the ejecta from the interstellar gas [8]. Furthermore, they also hold between any two points of a *smooth waveform* that is propagating steadily [9, 10].

The applicability of the RH relations and the resulting error depends on the degree of *unsteadiness*<sup>5</sup> of real shock waves. Generally, cylindrical and spherical shocks have a faster decay than plane shocks because of their geometric spreading. Furthermore, the unsteadiness of the  $P(t)$ -profile of air blast waves and underwater shock wave increases with decreasing distance from the explosive source and with increasing yield. Thus, micro shock waves (such as generated by powerful electric discharges, small chemical explosives or focused high-intensity laser radiation) have a higher degree of unsteadiness than astrophysical shock waves and terrestrial blast waves at large distances. Obviously, in reality there are always several deviations from ideal conditions made in the classic RH concept. Thus, the cardinal question arises how big this *overall systematic error*, caused by the application of non-ideal wave profiles, will be in comparison with an *overall measurement error*, caused by random errors and inaccurate calibration in the applied high-speed diagnostics.

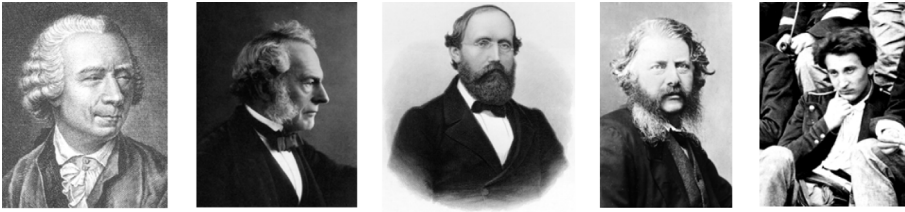
It is a difficult task to question the general validity of the RH relations, particularly in view of their good agreement with experimental results in gas dynamics. It seems that this nontrivial problem of error estimation has not yet been investigated in more detail. Up to this day, there exist no analytical proofs or numerical parametric studies, neither for justifying the numerous simplifications hitherto made in the original RH concept by a number of researchers nor for objecting to their use. It is very striking that, compared to the large number of publications on the application of the RH relations, the number of studies attempting to estimate the involved errors is extremely small – a surprising result also in regard to their wide applications beyond the shock physics community in the last seventy years.

Section 2 summarizes the RH relations in various coordinate systems, discusses their individual advantages for practical use and provides an RH terminology. Section 3 reviews and discusses published comments in the open literature on the admissibility of the RH relations. Section 4 provides a summary of modifications of the RH relations worked out since the early 1900s in S&E and illustrates the enormous spectrum on hand of practical examples covering shock propagation in all dimensions, both steady and unsteady. Section 5 compares ideal conditions made in the classical

---

<sup>4</sup> For a contact discontinuity across the surface  $P$  is continuous but  $\rho$ ,  $e$ ,  $T$  and  $s$  may be discontinuous, while for a shock  $P$ ,  $\rho$ ,  $e$ ,  $T$  and  $s$  are always discontinuous.

<sup>5</sup> There exists no definition of this term for a shock wave. Any unsteadiness will produce inconstancies of the fluxes of mass momentum and energy, thus violating the RH concept. Rise and decay times of the shock pressure pulse are certainly important parameters which influence the overall systematic error.



**Fig. 1.** Early contributors to the Theory of Conservation Laws in fluid dynamics: (a) Leonard Euler (1707–1783) [Dibner Library of the History of Science & Technology, Smithsonian Institution Libraries. National Museum of American History, Washington, DC]; (b) George G. Stokes (1819–1903) [The Royal Society of London]; (c) Bernhard Riemann (1826–1866) [R. Narasimhan (ed.): *Bernhard Riemann's gesammelte mathematische Werke, wissenschaftlicher Nachlass und Nachträge*. Springer-Verlag, Berlin (1990)]; (d) William J.M. Rankine (1820–1872) [Deutsches Museum, Munich]; and (e) Pierre-Henri Hugoniot (1851–1887) [Ecole Polytechnique, Paris].

RH relations with features existing in real shock waves, sample materials and some experimental techniques. Section 6 discusses the two most common cases of applying the RH relations to unsteady waves. Section 7 reviews published error estimations, discusses the nontrivial problem of error estimation in more detail and proposes a parametric study which might be adopted for quantitatively estimating systematic errors caused by a ramp-like rise and a linear or exponential decay of shock pressure. Section 8 is a summary of conclusions on the proper use of the RH relations and on problems arising in error data estimations.

For convenience to the reader, relevant citations from the RH literature are given below in full length. Supplementations in curved parentheses {...} are annotations made by the author.

## 2 Classical Rankine-Hugoniot relations

### 2.1 Brief early history

In 1757, Leonard Euler (Fig. 1a), a Swiss mathematician and then director of the mathematical class of the Berlin Academy in Prussia, published his famous paper on the motion of an inviscid liquid, gas etc. which can be applied to an incompressible as well as to a compressible flow [11]. In his analytical theory which is built upon the conservation of mass and momentum, he first applied partial differential equations to the problem – so-called *general Euler equations*. Introductorily he wrote: “However, sublime are the researches on fluids which we owe the Messr. Bernoulli, Clairaut and d’Alembert, they flow so naturally from my two general formulae that one cannot sufficiently admire this accord of their profound meditations with the simplicity of the principles from which I have drawn my two equations, and to which I was led immediately by the first axioms of mechanics”.

In 1848, Sir George G. Stokes (Fig. 1b), an Irish mathematical physicist, first considered discontinuous wave motion; i.e., the propagation of a shock wave [12]. He first correctly derived the jump conditions for conservation of mass and momentum, and concluded “equations (2) and (3) {later termed the *first two RH equations*} being satisfied, it appears that the discontinuous motion is dynamically possible”. However, he did not use the equation of conservation of energy. Stokes also first showed that the propagation of discontinuous pressure waves may occur with supersonic velocities.

In 1859, G.F. Bernhard Riemann (Fig. 1c), a famous German mathematician, presented his *Theory of Waves of Finite Amplitude* [13]. He applied the Euler equations on a one-dimensional (1D) shock wave propagating in an ideal gas, thereby assuming

an adiabatic reversible transition across the shock wave. Thus, he incorrectly assumed that the entropy remains unchanged through the shock transition – i.e., he conserved entropy rather than energy. His shock problem, defined as an initial value problem for a system based upon the laws of conservation, is known as the *Riemann problem*. It describes the time evolution of two constant states of an arbitrary shock parameter value,  $w_L$  and  $w_R$ , that are on the left and right of an initial discontinuity, respectively. He first recognized that the intersection of characteristic lines leads to a discontinuity (shock wave).

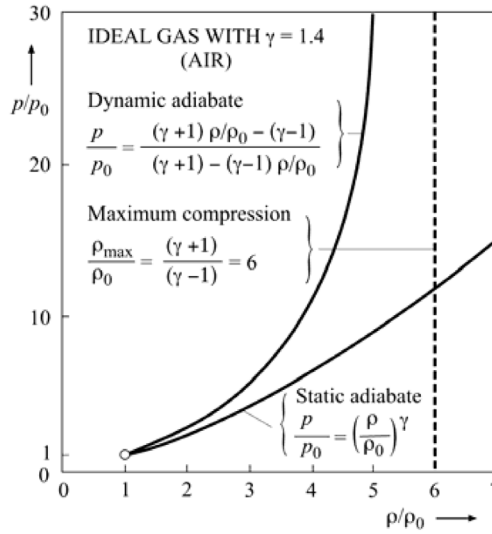
In 1869, William J.M. Rankine (Fig. 1d), a Scottish engineer, interested in the thermodynamics of wave propagation, presented a paper at the Royal Society of London “on *adiabatic waves*, that is waves of longitudinal disturbance in which there is no transfer of heat” and on the problem of “how to determine the relations that must exist between the laws of the elasticity and heat of any substance, gaseous, liquid or solid, and those of the *wave-like propagation of a finite longitudinal disturbance* in that substance” [14] – in modern terms, he treated the propagation of a *shock wave* in any medium. He concluded that the conservation laws of mass and momentum “hold for any type of disturbance, continuous or discontinuous, gradual or abrupt” and developed a condition for conservation of energy across a discontinuity in a fluid, assuming that across such a discontinuity a non-adiabatic process must occur rather than an adiabatic one. Rankine didn’t discuss the condition of reaching thermodynamic equilibrium. His third condition for conservation of energy is only equivalent to Hugoniot’s third equation for an ideal gas. Rankine also gave an explicit solution for the case of heat conduction and introduced the Greek letter  $\gamma$  to denote the ratio of specific heats which is still in current use.

In the mid-1880s, Pierre-Henri Hugoniot (Fig. 1e), a French military engineer and previous student of the famous Ecole Polytechnique at Paris, resumed the subject but obviously not knowing Rankine’s paper [15]<sup>6</sup>. Similarly to him, Hugoniot used in his theory the concept of conservation of mass, momentum and energy<sup>7</sup>. However, he first introduced the additional kinetic energy per unit mass of the fluid particles ( $=1/2u^2$ ) in the energy balance. Using Lagrangian coordinates and neglecting viscosity and heat conduction<sup>8</sup>, he showed that the passage of a shock through an ideal gas is not an isentropic process, but rather adiabatic and accompanied by an irreversible change of mechanical energy into thermal energy, thereby implying an increase of entropy across the shock. The concept of *entropy* was already introduced in 1865 by Rudolf J.E. Clausius (1822–1888), a German mathematical physicist [16]. However, the term was not used by Hugoniot in his 1889 paper.

<sup>6</sup> The manuscript of his epochal study on shock waves was entirely ready for print before his premature death during a business trip to the Compagnie des Tramways de Nantes. His manuscript was edited in 1887 by Roger Liouville (1856–1930), a young French mathematician.

<sup>7</sup> Rankine [14] called the *internal energy* the “work done in altering bulkiness” and the *kinetic energy* the “actual energy of the disturbance”, the latter was called by Hugoniot “la force vive correspondant au mouvement sensible”, a quantity for a gas given by him by  $P/[\rho(\gamma-1)]$ . Concerning the internal energy, Hugoniot [15] wrote: “La quantité sera désignée simplement sous le nom d’énergie interne, faisant ainsi abstraction de la portion de cette énergie qui ne pourrait être utilisée sans modification dans la constitution du gaz... On sait que, dans toute transformation subie par un élément de masse d’un gaz sans gain ni perte de chaleur, le travail des forces extérieures est égal à l’augmentation de l’énergie totale, c’est-à-dire de la somme de l’énergie interne et de la force vive”.

<sup>8</sup> In 1910, Sir Geoffrey I. Taylor [83], a British physicist and applied mathematician, had first worked out an explicit formula for weak shocks which considered both viscosity and heat conduction.



**Fig. 2.** Comparison between the “static adiabetic” (or “Poisson adiabetic”) and the “shock adiabetic” (or “Hugoniot”) which in the mid-1880s Pierre-Henri Hugoniot derived for an ideal gas with  $\gamma = 1.4$  (air), thereby assuming a step function in time for all shock parameter values. Note that it requires a higher pressure to compress a gas to a given density ratio  $\rho/\rho_0$  using a single shock wave than by static compression [17], p. 150.

Hugoniot had derived the dynamic adiabetic for an ideal shock-compressed gas which, in modern notation, is given in terms of the pressure by

$$P_H = P_0[\zeta\rho_H/\rho_0 - 1]/[\zeta - \rho_H/\rho_0] \quad (1a)$$

or in terms of the density by

$$\rho_H = \rho_0[1 + \zeta P_H/P_0]/[\zeta + P_H/P_0], \quad (1b)$$

where  $\zeta = (\gamma + 1)/(\gamma - 1)$  and  $P_0$  and  $\rho_0$  are the atmospheric pressure and density, respectively. Equation (1a), today bearing his name, is steeper in the pressure-density plane than the “Poisson adiabetic” or “static adiabetic” (Fig. 2). He also found that for a perfect gas the maximum compression is given by

$$(\rho/\rho_0)_{\max} = (\gamma + 1)/(\gamma - 1), \quad (2)$$

where  $\gamma$  is the ratio of specific heats. For air ( $\gamma = 1.4$ ) this results in a maximum compression of (only) six.

Biographical and bibliographical data on Euler, Hugoniot, Rankine, Riemann, and Stokes, also including relevant secondary literature, have been given elsewhere [17]. The early history of the RH relations was also discussed in the international journal *Shock Waves* [18–20].

It is interesting here to note that at the time when Stokes studied the properties of shock waves from the mathematical point of view, their existence was not yet demonstrated experimentally. When Rankine published his paper in 1869, the German physicist August Toepler had already five years before first visualized subjectively the discontinuous nature of an aerial shock wave using a stroboscopic arrangement. However, it is rather unlikely that this pioneering achievement, first published in Toepler’s Ph.D. thesis in 1864, was known to Rankine and Hugoniot. The famous

first photograph of a shock wave, the bow shock surrounding a supersonic projectile, was first published by the Austrian physicists Ernst Mach and Peter Salcher in 1887, Hugoniot's year of death [17].

## 2.2 RH Relations in different coordinate systems

The classical 1D RH relations provide information on the important shock parameter values and are widely used in nonlinear acoustics, gas dynamics, supersonic aerodynamics, and condensed-matter shock physics. These equations can be derived

- (i) by considering that mass, momentum and energy are conserved across the shock, a straightforward manner which can be found in many textbooks [21, 22];
- (ii) from the Euler conservation equations [1, 23–26]. In the frame of a steady plane shock, both upstream and downstream states of the medium are assumed to satisfy the Euler equations in differential form. Ideally, they should be integrated from negative infinity to the shock for the upstream and from the shock to positive infinity for the downstream. Often, for a wide enough upstream and downstream region, the reasonable steady state approximations are then the RH relations given below; and
- (iii) from the theory of hyperbolic conservation laws [27]<sup>9</sup>.

---

<sup>9</sup> For a 1D inviscid hyperbolic system in the conservation form

$$dw/dt + df(w)/dx = 0, \quad (14)$$

it can be shown that the solution is allowed to be discontinuous and that a simple relation, termed *RH condition*, exists between a scalar conserved quantity or state variable  $w(x, t)$ , a flux function  $f(w)$  and a scalar quantity  $\lambda = dx/dt$ . The speed of the discontinuity or shock along a trajectory  $x(t)$  is given by

$$\lambda = [f(w)]/[w], \quad (15)$$

where  $[\cdot]$  indicates a jump in some quantity across the discontinuity; i.e.,  $[\text{speed of discontinuity}] = [\text{jump in the flux}]/[\text{jump in the state}]$ . In the example of a shock wave propagating in the Eulerian frame with speed  $\lambda = U_s$ , the jump in the state from “0” at  $x < \lambda t$  to “1” at  $x > \lambda t$  is given by

$$[w] = w(x(t)_1, t) - w(x(t)_0, t), \quad (16)$$

and the corresponding jump in the flux by

$$[f(w)] = f\{w(x(t)_1, t)\} - f\{w(x(t)_0, t)\}. \quad (17)$$

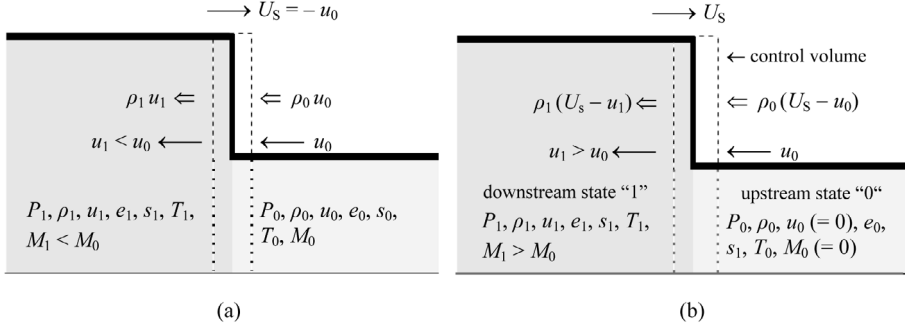
The first two RH relations are then given by

$$\begin{aligned} \lambda[\rho] &= [\rho w]; & \Rightarrow \lambda(\rho_1 - \rho_0) &= \rho_1 u_1 - \rho_0 u_0; \\ \lambda[\rho w] &= [\rho w^2 + P]; & \Rightarrow \lambda(\rho_1 u_1 - \rho_0 u_0) &= (\rho_1 u_1^2 + P_1) - (\rho_0 u_0^2 + P_0). \end{aligned}$$

In order to put the energy equation into conservation form, it is necessary to use the energy equation in terms of the total specific enthalpy given by  $H = 1/2u^2 + e + P/\rho$ , where  $e$  is the specific internal energy, hence

$$\lambda[\rho H] = [\rho u H]; \quad \Rightarrow \lambda(\rho_1 H_1 - \rho_0 H_0) = H_1 \rho_1 u_1 - H_0 \rho_0 u_0.$$

In a Lagrangian frame the shock is stationary ( $\lambda = 0$ ), while in an Eulerian frame the shock propagates with the shock front velocity ( $\lambda = U_s$ ). Note that, under the usual assumption of a constant shock velocity  $\lambda$  (hence  $d\lambda/dt = 0$ ), equation (14) is only valid for a constant flux function, hence  $df(w)/dx = 0$ .



**Fig. 3.** Parameter values at both sides of a shock front, illustrated at the examples of (a) Lagrangian coordinates, an observer riding on the shock and (b) Eulerian coordinates, an observer seeing the shock from a fixed position. The RH concept implies that the flow is supersonic ahead of the shock wave and subsonic behind; the reverse possibility is ruled out by the requirements of non-decreasing entropy. The width of a control volume enclosing the discontinuity is very small (in the nm-range) compared to the characteristic length of the system (in the cm- to m-range). Usually, after passing the shock front all shock parameter values jump up, but in the Lagrangian frame two shock parameter values jump down ( $u_1 < u_0$  and  $M_1 < M_0$ ).

The three conservation equations which require continuity of the fluxes of mass, momentum and energy are given here for both coordinates in modern notation.

### 2.2.1 Symbols

Throughout this paper, the flow variables denote pressure  $P$ , density  $\rho$ , specific volume  $v = 1/\rho$ , particle velocity  $u$ , internal energy  $e$  per unit mass (both thermal and elastic), temperature  $T$ , entropy  $s$ , and Mach number  $M$  of a steady planar shock. Referring to subscripts “0” (upstream) and “1” (downstream), they correspond to “pre-shock” and “post-shock” conditions, respectively.  $U_s$  is the shock front velocity in the Eulerian frame,  $\gamma$  is the ratio of specific heats.  $\tau$  and  $\Theta$  denote the rise-time and time constant of initial decay, respectively, of an unsteady wave with pressure profile  $P(t)$ . The individual relative errors  $\eta_1, \eta_2, \dots, \eta_{20} \dots$  are due to violations of the RH concept (see Sects. #1–#20).  $\varepsilon$ ,  $\eta$  and  $\Delta$  are measurement, systematic, and total errors, respectively which are based upon the RMS (Root Mean Square) measure.

### 2.2.2 Lagrangian or material coordinates

These particle-tracking coordinates (Fig. 3a) apply a reference frame which, moving with the shock wave velocity, is given by  $u_0 = -U_s$ . A Lagrangian system considers temporal positions of each flow particle, thus providing information on their history.

The flow on one side of the shock is specified by the velocity  $u_0$ , the density  $\rho_0$ , the pressure  $P_0$ , and the specific internal energy  $e_0$  [erg/g] with corresponding values of  $u_1$ ,  $\rho_1$ ,  $P_1$ , and  $e_1$  on the other side:

conservation of mass:

$$\rho_0 u_0 = \rho_1 u_1; \quad (3a)$$

conservation of linear momentum:

$$P_0 + \rho_0 u_0^2 = P_1 + \rho_1 u_1^2; \quad (4a)$$

conservation of energy:

$$e_0 + 1/2 u_0^2 + P_0/\rho_0 = e_1 + 1/2 u_1^2 + P_1/\rho_1.$$

Eliminating the two velocities, one arrives at the *Hugoniot equation*

$$e_1 - e_0 = (P_0 + P_1)(1/\rho_0 - 1/\rho_1). \quad (5a)$$

From equations (3a) and (4a) the following relation can be worked out:

$$P_1 - P_0 = (-u_0/v_0)^2(v_0 - v_1) = (U_s/v_0)^2(v_0 - v_1) = U_s^2 \rho_0^2 (1/\rho_0 - 1/\rho_1). \quad (6a)$$

Historically, Lagrangian coordinates were used in gas dynamics in numerous pioneering papers on shock waves. Famous examples include the classical solution of a high-intense point explosion into a uniform medium, carried out in connection with the first atomic explosion at Trinity Site, NM on June 16, 1945 [28], and pioneering studies in condensed-matter shock wave physics for the measurement of dynamic EOS data using various experimental techniques which applied planar shock waves of controllable amplitude.

Lagrangian coordinates may also simplify the analysis of unsteady shock phenomena such as observed in wind tunnels and in the interior of laboratory and astrophysical supersonic jets. Important diagnostic techniques of recording particle velocities in the Lagrangian frame include optical interferometer techniques [29], Doppler shift measurement of resonances resulting from moderate neutrons passing through the sample [30], “in-material” gauges [31], and electrical contact pins [32].

### 2.2.3 Eulerian or laboratory coordinates

They are fixed in space (“laboratory frame”), therefore are coordinates describing the instantaneous position of an element of fluid (Fig. 3b). In modern notation they are for conservation of mass:

$$(U_s - u_1)\rho_1 = (U_s - u_0)\rho_0; \quad (3b)$$

conservation of linear momentum:

$$P_1 - P_0 = \rho_0(U_s - u_0)(u_1 - u_0); \quad (4b)$$

conservation of energy:

The net work done by  $P_1$  and  $P_0$  is given by [21]

$$P_1 u_1 - P_0 u_0 = \rho_0(U_s - u_0)[e_1 - e_0 + 1/2(u_1^2 - u_0^2)].$$

In the case of an undisturbed fluid ( $u_0 = 0$ ) and with equations (3b) and (4b) the following relations can be worked out:

$$e_1 - e_0 = (P_0 + P_1)(1/\rho_0 - 1/\rho_1). \quad (5b)$$

$$P_1 - P_0 = (U_s/v_0)^2(v_0 - v_1) = (U_s \rho_0)^2(1/\rho_0 - 1/\rho_1). \quad (6b)$$

In a  $(P, v)$ -diagram equation (6b) represents a straight line of negative slope joining the initial and final states which is proportional to  $U_s^2$ , usually designated as **R**( $p, v$ ) and called *Rayleigh line* in the western world and *Mikhel'son line* in Russia<sup>10</sup>.

---

<sup>10</sup> For a true step wave, characterized by a step rise and steady propagation, pressure loading from  $P_0$  to  $P_1$  occurs as a jump in zero time. In real shock waves with a finite rise time,  $P_1$  is presumably reached along a slightly curved rather than a straight line, because the classical RH relations are no longer valid.

Eulerian coordinates are useful in the analysis of shock tube operation parameter values; blast parameter values resulting from chemical and nuclear explosions in the atmosphere or underwater; sonic booms driven by supersonic projectiles and aircraft in the atmosphere (*Mach bow shock*); shock waves coupled into a solid by an adjacent detonation wave (*von Schmidt head wave*); and stellar bow shocks resulting from the interaction of a stellar wind with a planet's magnetopause (*Earth's bow shock*). Eulerian-frame shock parameter values can be measured using optical high-speed photography (e.g. reflected-light photography, schlieren, shadowgraphy, optical interferometry), flash radiography, particle trajectory techniques, and other high-speed diagnostic methods such as pressure and strain gauges.

#### 2.2.4 Hybrid or mixed coordinates

An Eulerian-Lagrangian coordinate system utilizes the Euler form of the RH equations, but introduces Lagrangian coordinates. It has been proved useful, apparently for the first time, for calculating the pressure wave originating from a spherical charge underwater [33]. This *Brinkley-Kirkwood theory* uses the equations of motion and continuity in mixed coordinates specialized at the shock and the momentum equation of the RH relations in a differentiated form such that the shock front is steady. This theory has also been used and extended by astrophysicists to study shock locus and shock decay of stellar explosions.

*Arbitrary Lagrangian-Eulerian (ALE)* numerical codes allow material to flow through the mesh; they apply a distorted Lagrangian mesh and a spatially fixed Eulerian mesh [34]. For general hydrodynamic problems involving large deformations or moving structures, the full Navier-Stokes equations need to be solved in a moving mesh, since the fluid is described in a moving domain because of the structure motion. In a more recently published book weak and strong coupling problems and their solutions using ALE methods are discussed [35].

#### 2.2.5 Annotations

- (i) In the derivation of the RH relations a mathematical trick is applied [1, 2]: the shock of instantaneous position  $sw(t)$  is considered as an “internal boundary” and the shock parameter values are distinguished on both sides of the shock by  $sw(t) - \xi$  and  $sw(t) + \xi$ , where  $\xi > 0$  is a small quantity and independent of time. The shock width  $sw(t)$  is supposed to be contained in an interval  $(a, b)$  ranging from  $a \rightarrow sw(t) - \xi$  to  $b \rightarrow sw(t) + \xi$ , where  $a$  and  $b$  are arbitrary coordinates independent of time. Under consideration that  $a$  approaches the upstream side and  $b$  the downstream side of the shock,  $sw(t)^-$  and  $sw(t)^+$ , respectively; i.e., taking the limit  $a \rightarrow sw(t)^-$  and  $b \rightarrow sw(t)^+$ , it is possible to differentiate under the integrals of the conservation law using the Leibniz rule. Thus, the resulting RH relations hold for an infinitesimal shock rise-time. A finite shock rise-time would provoke that the material is compressed along a dynamic adiabat off the Hugoniot (see Sect. 7.4).
- (ii) Note that in the Lagrangian frame  $u_0 > u_1$ ,  $M_0 = u_0/c_0 \geq 1$ , and  $M_1 = u_1/c_0 \leq 1$ , while in the Eulerian frame  $u_0 < u_1$ ,  $M_0 = u_0/c_0 \leq 1$ , and  $M_1 = U_s/c_0 \geq 1$ . Here  $M_0$  is the upstream Mach number,  $M_1$  is the downstream Mach number, and  $c_0$  is the *sound speed* in a gaseous or liquid material or the *bulk sound speed* in a solid material.
- (iii) In most cases the shock propagates into a medium at rest (in the Eulerian frame:  $u_0 = 0$ ) and of known initial density  $\rho_0$ . Note that in condensed-matter shock physics  $P_1 \gg P_0$ .

- (iv) The increase of the specific entropy [J/kg K] of the gas across the shock is given by

$$\Delta s = s_1 - s_0 = R/(\gamma - 1) \times \ln[P_1/P_0 \times (\rho_1/\rho_0)^\gamma], \quad (7)$$

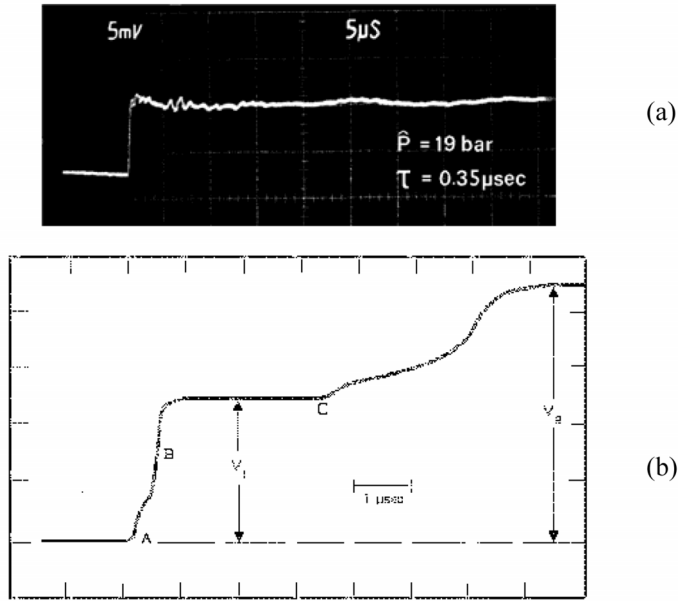
where  $R$  is the gas constant per unit mass. The temperature jump at the shock front is given by

$$\Delta T = T_1 - T_0 = T_0[(P_1/P_0)(\rho_0/\rho_1) - 1], \quad (8)$$

where  $T_0$  and  $s_0$  are the temperature and entropy of the quiescent gas, respectively. Note that equations (5a) and (5b) are independent of the chosen coordinate system.

- (v) In the shock compression of solids, pressures  $P_0$  and  $P_1$  must be replaced by stresses  $\sigma_0$  and  $\sigma_1$ , respectively, which denote the total stresses in the direction of shock compression.
- (vi) To precisely determine Hugoniot curves of homogeneous materials, different experimental techniques have been developed which approach step functions with low rise-time of two shock parameter values. These values are then measured a certain time behind the shock front to allow for reaching thermodynamic equilibrium in the shock-compressed sample material, for example in gas dynamics by measurement of  $P(t)$  and  $U_s$  in a shock tube [36] (Fig. 4a) and calculating  $\rho_1(t)$  and  $u_1(t)$  using equations (3b), and (4b); and in condensed-matter shock physics by measurement of  $u_0$  and  $u_1$  using a plate-impact experiment (Fig. 4b) and calculating  $P_1(\rho_1)$  using equations (3a) and (4a). Note that this procedure provides only a *single point* on the Hugoniot curve. In order to get a complete Hugoniot curve, a considerable number of such experiments are required using step-like shock parameter values of different amplitudes.
- (vii) For special applications it might be helpful to use either the integral form or the differential form of the conservation laws [37]. The former implies that the conservation laws are satisfied for a control volume while the latter means that the laws are satisfied everywhere in the flow field [38, 39]. The integral form has been considered as being more fundamental than the differential equation providing the basis for the mathematical theory of weak solutions and including the derivation of the RH relations that govern the shape and speed of shock waves.
- (viii) From the RH relations many useful relations for normal shock waves can be derived in terms of either the upstream Mach number or the shock strength ( $P_1/P_0 > 1$ ) across the shock [40]. In Canada, *RH Spreadsheets* have also been worked out relating 28 different physical properties behind a shock moving into an almost ideal gas such as air (with  $\gamma = 1.4$ ) at normal temperature (15 °C) and normal pressure (101 325 Pa) [41].
- (ix) A number of comprehensive *Shock Hugoniot Data Banks* have been established over the last decades, covering gases, gaseous mixtures, liquefied gases, rarefied gases, organic and inorganic liquids, most solid elements, porous solids, chemical compounds, minerals, rocks and mixtures of minerals, polymers, different kinds of wood, ice and snow of different compaction, and even explosive gaseous mixtures, liquid and solid explosives [42–49]. Hugoniot (or shock adiabatics), usually given for real gases as  $P_H$  ( $\rho_H$ ,  $T_H$ ), and for liquids and solids<sup>11</sup> as  $P_H$  ( $\rho_H$ ,  $e_H$ ) data, are very useful for materials researchers and military engineers, geologists, geophysicists, planetary scientists and astrophysicists in data analysis and numerical simulations of shock-loaded materials. They also helped theoreticians to better understand the compression of matter on the molecular level and

<sup>11</sup> Strictly seen, at high stresses  $\sigma_H$  normal to the wave front rather than at high pressures  $P_H$ .



**Fig. 4.** Examples of a *quasi-steady approach* for applying the RH relations to unsteady waves with a finite rise-time and temporarily limited constant fluxes: (a)  $P(t)$  profile of an aerial shock wave with a rise-time of  $\sim 350$  ns measured with a piezoresistive carbon gauge which, compared to piezoelectric pressure gauges, has a very short rise of only some 10 ns (after [204]); (b) continuous oscilloscope monitoring of “in-material” particle velocity during compression and subsequent release in a geologic solid geologic material (Solnhofen limestone) produced by plane impact from a light gas gun. An embedded electromagnetic gauge was used as a “Lagrangian” probe. The particle velocity  $u_1(t)$  was converted by the gauge into a voltage signal  $V(t)$  which shows the shock rise-time, the almost constant particle velocity of amplitude  $V_1$  and the rarefaction wave of amplitude  $V_2$ . Note that the rise-time of  $\sim 500$  ns is followed by an almost horizontal plateau with duration of  $\sim 3100$  ns (after Ref. [82]).

to formulate dynamic equations of state. The establishments of Hugoniot data banks, mostly initiated by governmentally supported projects which provided research institutes with enormous financial support over decades, stimulated the development of new dynamic compression methods, ranging from the early use of high explosives in direct contact with the sample up to more sophisticated devices such as gun-driven impactors, explosively-driven flying plate methods up to high-intensity pulsed laser beams in modern times. Furthermore, they promoted the invention of numerous new instrumental and diagnostic techniques with the benefits to better solving problems also in other disciplines of S&E.

### 2.3 RH Terminology

In the western literature different terms are used for equations (3)–(5): *classical* (or *standard*) *RH relations*, *RH jump conditions for shocks*, *RH discontinuity conditions*, *RH shock relations*, *RH conservation relations*, *RH equations*, or *shock relations*. The *inverse RH relations* are used in gas dynamics, particularly for characterizing blast waves, and provide the shock Mach number as functions of the physical properties across a shock [50].

Equations (3) and (4) were initially named the *Riemann equations* [51]. But they were first given by Stokes [12]. In the former Soviet Union, the term *RH relations* for the first two conservation equations was not adopted by leading Soviet shock physicists, some rather (correctly) referred to Stokes. But they adopted the term *Hugoniot* for the Hugoniot curve.

Equation (5a or 5b) is the *Hugoniot equation* (“Hugoniot-Gleichung”), originally termed by the German theoretical physicist Richard Becker [52]. But also the terms *Hugoniot law* and *shock adiabetic* are used. Note that the term *Hugoniot law* also has another meaning in fluid dynamics [53]<sup>12</sup>.

In a compressible flow, plots of  $P_H(\rho_H)$  and  $P_H(v_H = 1/\rho_H)$ , “centered” at  $(P_0, \rho_0)$  and  $(P_0, v_0)$ , respectively, are known as *Rankine-Hugoniot (RH) curves*, *Hugoniot curves* or briefly as *Hugoniot*s. For all normal fluids, the Hugoniot  $P_H(\rho_H)$  is a concave curve upward. Thus, the slope  $dp_H/d\rho_H$  – and therefore, the sound velocity  $c = (dp_H/d\rho_H)^{1/2}$  – increases with increasing density  $\rho_H$ .

The term *RH function* relates to  $P_H = f(v_H, e)$ . In the literature the term *Hugoniot adiabetic* is also used for the shock adiabetic,  $P_H/P_0 = f(\rho_H/\rho_0)$ , to distinguish from the *Poisson adiabetic*,  $P_1/P_0 = (\rho_1/\rho_0)^\gamma$  [54].

The *RH condition*, equation (15)<sup>9</sup>, is a criterion which a shock  $x(t)$  must satisfy. It expresses conservation of a shock parameter value  $w$  across jump discontinuities.

The *RH problem* is related to planetary bow shocks and interplanetary shocks. It consists of taking the spacecraft observations (or time series) of density, velocity and magnetic field across the Earth’s bow shock in order to determine if the discontinuity is indeed a shock and of what class – a nontrivial problem which involves the specification of ten variables [55].

*RH shocks* are classical shock waves for which the shock wave parameter values are related to each other by the RH relations. In RH shocks no energy is radiated away through the surface of the flow; i.e., the radiative cooling mechanism is extremely inefficient and energy is conserved [56]. The term has been used by astrophysicists in connection with studies on standing shocks in viscous disc-like accretion flows around black holes.

*Non-RH shock waves* don’t satisfy the RH jump conditions such as discontinuity phenomena observed in reflected or branched shocks (*Mach effect*) [57].

### 3 Comments in the literature on the applicability of the RH conditions

In the literature, the spectrum of opinions on the admissibility of the RH relations to real shock waves is complex and somewhat contradictorily, ranging from unrestricted approval to critical reservation. Hence, there is a need for a review and a more detailed discussion.

<sup>12</sup> There it denotes a relation in a tube of a 1D flow between cross-section  $F$ , velocity  $C$  of a compressible fluid, sound velocity  $a$ , and Mach number  $M$  given by

$$dF/F = dC/C \times (C^2/a^2 - 1) = dC/C \times (M^2 - 1).$$

### 3.1 Comments on unrestricted applicability

In 1948, Robert H. Cole, a US physical chemist at Brown University (Providence, RI), calculated shock parameters of *underwater shock waves*<sup>13</sup> using the RH relations [21]. He took the view that “these relations, being obtained for the limit of negligible thickness of the shock front, should be equally valid for plane or spherical waves” [58].

In 1955, John W. Walsh and Russell H. Christian, two eminent pioneers of condensed-matter shock physics at Los Alamos (NM), considered the experimental difficulties to obtain reliable Hugoniot data as more severe than problems with using the RH relations in their analysis of experimental data [59]. They concluded that “the application of the RH equations is exact, of course, regardless of the thickness of the shock front. The limitation is imposed because of the finite scale of experimentation”.

In 1958, Melvin H. Rice and collaborators at Los Alamos reviewed current shock production methods and measurement techniques for determining pressure-compression states behind shock waves [60]. They annotated that the RH relations “are independent of the thickness of the shock front, provided only that the shock front pressure profile is independent of time {i.e., assuming a steady wave}”.

In 1962, Newby Curle, a British fluid dynamicist at the University of Southampton, using Lagrangian coordinates and developing the RH relations in the usual way, stated that “when the shock is not stationary, all these {RH} equations are still true provided that the flow velocities  $u_1$ ,  $u_2$  {here  $u_0$ ,  $u_1$ } are measured relative to the shock” [61].

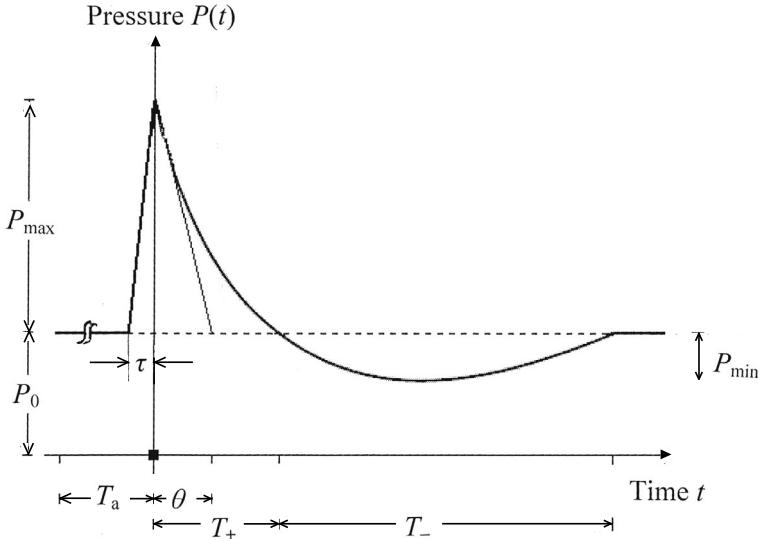
In 1964, Samuel Glasstone, a British-born US scientist who together with Philip J. Dolan compiled and edited the famous book “The Effect of Nuclear Weapons”, wrote: “The foregoing results {on the properties of blast waves} have referred to blast wave conditions which are ideal or nearly ideal, so the RH equations are applicable” [62]. An ideal blast, originated from an explosion in air, is characterized by an almost instantaneous increase of the ambient pressure followed by a prompt decay. Mathematically, the blast wave profile  $P(t)$  can be described by a *Friedlander function*, see equation (9) and Figure 5.

In 1971, John W. Dewey, a physics professor at the University of Victoria (BC, Canada), studied the propagation of blast waves originated from large hemispherical TNT surface explosions [63]. In 2006, Dewey transformed the RH equations in order to describe the physical properties of the gas behind a blast wave, moving into a stationary ambient atmosphere, in terms of the Mach number of the shock and concluded: “Assuming the ratio of specific heats of air to be 1.4, the RH equations provide exact values of the physical properties behind the primary shock of a blast wave, with accuracy such that this is the preferred method to calibrate electronic transducers” [50]. In a more recent private communication, Dewey wrote that he had tested the validity of the RH equations many times and had never found to be at fault [64].

In 1983, Indian shock physicists at the Bhabha Atomic Research Centre (Bombay, India), discussed equation-of-state theories of condensed matter up to ultra-high pressures, thereby also addressing the applicability of the RH relations [65]. They wrote “These relations are valid at the shock front even for the unsteady shock wave (where the stress behind the shock front is not uniform) and are independent of the curvature of the shock front”.

In 1988, Lawrence E. Murr, an engineering professor at the Oregon Graduate Center (Portland, OR), and his collaborator argued that the familiar RH relations “are equally valid for plane and spherical shock waves, {however, they} do not include

<sup>13</sup> Underwater shock waves resulting from an underwater explosion of a TNT charge are unsteady waves, because the pressure roughly decreases with time proportional to about  $e^{-t/T}$ , where  $T$  is the time constant of decay.



**Fig. 5.** Example of a *pseudo-steady approach* for applying the RH relations to an unsteady wave with a shock rise-time  $\tau$  and decaying fluxes: the pressure-time profile of an air blast wave can be described by a modified Friedlander function, equation (9). Here  $P_0$  is the ambient pressure,  $P_{\max}$  the maximum overpressure in the positive phase  $T_+$ ,  $P_{\min}$  the minimum pressure in the negative phase  $T_-$ ,  $\tau$  the blast rise-time (usually  $\tau \ll T_+$ ), and  $\theta$  the time constant of initial decay ( $\theta = 0.5T_+$ ). Note that at time  $\delta = 0.01T_+$ , a time period which is very large compared to  $\tau$  and after which thermodynamic equilibrium has been reached, the pressure has decayed to only 98.01% of  $P_{\max}$ . Thus, the assumption of a constant flux behind the shock front is a good approximation and the application of the RH relations seems to be justified. Note that for a proper recording of blast waves pressure gauges with short rise-times are indispensable for measuring the true peak pressure  $P_{\max}$ .

heat conduction or similar considerations” [66]. Concerning the applicability of the RH relations they also annotated that “instabilities in crystalline materials will create dislocations and other crystal defects within the shock front as it propagates”.

In 2009, researchers at the Laboratory for Laser Energetics of the University of Rochester (Rochester, NY) carried out precise measurements of the equations of state of polystyrene and polypropylene using laser-driven shock waves with pressures ranging from 1 to 10 Mbar [67]. In their analysis of the *impedance matching technique* – a method which is used to infer the particle velocity by referencing the sample under study to a standard material whose EOS is known [59, 68] – they also used the first two RH relations and annotated: “The shock jump conditions are a consequence of conservation of mass and momentum, which are always satisfied regardless of shock stability. Therefore, shock steadiness is not a requirement for the impedance matching technique with transparent standards, since jump conditions for a shock hold for decaying (and increasing) shock waves. If the measurement has sufficient time resolution, the requirement for shock steadiness can be relaxed as long as the variation in velocity can be measured”.

In 2013, Charles E. Needham, a US physicist at Applied Research Associates (Albuquerque, NM) and author of the book “Blast Waves”, annotated to the author: “The RH relations can be derived directly from the Euler conservation equations. Because the Euler equations are differential equations, and the shock is assumed to be discontinuous, the conditions for the RH relations hold at infinitesimal distances and times on either side of the shock. RH relation derivation does not require a square

wave, but it is easier to demonstrate to students than to derive from partial differential equations” [69].

### 3.2 Comments on restricted applicability

In 1950, John von Neumann, a renowned mathematician at Institute for Advanced Study (Princeton, NJ), and Robert D. Richtmyer, a Los Alamos physicist, annotated that in order to numerically calculate hydrodynamic shocks it requires that “the RH equations are satisfied, provided the thickness of the shock layers is small in comparison with other physically relevant dimensions of the system” [70]. Using their famous concept of an artificial nonlinear shock viscosity they showed that, applied on an ideal gas with viscosity, the RH relations were satisfied provided in the upstream/downstream flow an equivalent “viscous pressure” is small or zero far away from the shock. Their shock profile was sinusoidal and approximately six elements wide.

In 1959, Harold L. Brode, a US nuclear weapons effects physicist at The RAND Corporation (Santa Monica, CA), found that the blast in real air differs from that in an ideal gas (as assumed in the relations worked out by Rankine and Hugoniot with  $\gamma = 1.4$ ) in a number of important ways [71]. His numerical studies had shown that the air blast is only 60% as efficient as the ideal gas blast insofar as peak overpressures are concerned, since the various dissociations and ionizations of molecules and atoms in high-temperature air near the source tend to reduce the blast efficiency in real air.

In 1961, William Band at Washington State University (WSU, Pullman, WA) and George E. Duvall at SRI’s Poulter Laboratory (Menlo Park, CA), two US condensed-matter shock physicists, presented simple derivations of the RH relations [72]. It is well-known that the profile of any large disturbance will change during propagation, eventually developing the front of a propagating wave into a discontinuity. A steady propagation of an unchanging profile – already studied by Lord Rayleigh [73] and named by him the “permanent regime profile” – is possible in a viscous, heat-conducting fluid which has to be chosen precisely to fit the equations of motion [74]. Band and Duvall discussed the (practically important) case that the shock is not a discontinuity but rather characterized by a rapid increase from an initial undisturbed value at the foot of the shock up to a *peak value* at the crest, followed by a rarefaction. They stated that “if this profile is maintained with time, we can choose a new reference system in which the shock remains permanently at rest, and it is possible to apply the three conservation theorems”. They derived the three RH relations in an Eulerian frame in the usual way, but considered the particular case where the point of interest is at the crest of the shock, so that  $u_1$ ,  $e_1$ ,  $P_1$ , and  $\rho_1$  refer to peak values. They wrote, “The relation between peak stress and peak compression so obtained for shocks of various amplitudes is called the *Hugoniot equation of state* of the material. It represents the locus of all points which can be reached by shock compression from the initial state  $P_0$ ,  $\rho_0$ ”.

In 1969, George E. Duvall, an applied physics professor at WSU and dean of US shock physics, presented in Italy at the International School of Physics Enrico Fermi a paper on shock waves in condensed media. He annotated that “experimentally produced shock waves seldom exactly satisfy the requirements of steady flow assumed in deriving the jump conditions. The states connected by the shock transition may not be precisely uniform or the shock wave has not propagated far enough to become steady. However, the experimental conditions may be very close to the theoretical assumptions, and it is quite likely that errors involved in applying the jump conditions to experiments are less than those originating from other sources. Such errors may be significant if gradients in adjacent regions are comparable to those in the shock or if time has not been sufficient for the flow to become steady and the curvature of the

Hugoniot is large. The resolution of this question is a constant source of concern to experimentalists and no satisfactory resolution has been made...” [75].

In 1970, G. Richard Fowles at WSU and Roger F. Williams at SRI’s Poulter Laboratory, two US condensed-matter shock physicists, published an analytical extension of the RH relations for arbitrary *unsteady plane shock waves* in which equilibrium or steady state may not yet be obtained [76, 77]. Their unusual analysis is based on the measurement of wave profiles at several positions in a shock-compressed sample. They wrote: “Some experimenters have applied the RH jump conditions incrementally to each part of an observed, unsteady wave; this treatment is only partially correct, as we show in Section 2 {where they showed what measurements are necessary and how they should be interpreted in order to deduce directly the associated values of stress, density, and internal energy for each portion of an observed shock}... We now ask what measurements of wave properties (that are experimentally feasible) are required in order to be able to deduce the corresponding values of pressure, specific volume and energy associated with each part of an observed wave”. They also discussed how dynamical histories might be deduced from measured histories, independent of assumptions concerning *material constitutive relations*<sup>14</sup>. In the same year, Fowles also extended the conservation relations to unsteady cylindrical and spherical shock waves [78].

In 1971, shock physicists at SRI’s Poulter Laboratory using the Fowles-Williams method [76] first calculated the constitutive relationships from multiple *Lagrangian gauge*<sup>15</sup> records obtained in unsteady 1D shock wave experiments by integrating the flow equations which express conservation of mass and momentum along a particle path [79]. However, their method of combined measurements of stress and particle velocity – in the following years modified by SRI physicist Lynn Seaman [80] who analyzed either longitudinal stress or longitudinal particle velocity alone – is difficult to implement experimentally. Later, physicists at WSU, primarily considering the analysis of particle velocity histories, extended these investigations by using surfaces explicitly fit to measured histories and directly integrating the conservation equations [81].

In 1971, shock physicist William Murri and some of his colleagues at SRI’s Poulter Laboratory, while reviewing recent advances in condensed-matter shock wave physics, emphasized the change in viewpoint that has been brought about by the appearance of in-material gauges of high temporal resolution [82]. Concerning the applicability of the RH relations, they made the following interesting comment: “The RH conditions hold only for *steady waves* or for *truly discontinuous jumps* in stress, density and particle velocity. Intuitively one might expect that truly steady shock waves rarely occur, and this is in fact the case. The conditions under which steady shock waves may occur were originally discussed by Lord Rayleigh [73] and G.I. Taylor [83], and a recent review of subsequent work has been given by Kelly and Gillis [84]. Thus, the RH jump conditions only rarely rigorously apply”.

In 1972, R.T. Walsh, a materials scientist at Sandia Laboratories (Albuquerque, NM), wrote: “As a result of the varying shock width, the RH conditions are not exactly satisfied in the case of a decaying shock... Aside from uncertainties in the material properties, this is probably the largest single source of error in one-dimensional solid

<sup>14</sup> Duvall and Graham gave the following definition of this term: “The material description is stated as a set of constitutive relations. In the simplest case this set is the equation of state alone... More generally, it includes equations describing various irreversible and rate-dependent processes important to the problem at hand...” [86].

<sup>15</sup> A *Lagrangian gauge*, an electromagnetic gauge based on the principle of a moving conductor in a magnetic field and embedded in the sample material, moves with the material during compression. Thus, it provides profiles of particle velocity vs. time.

mechanics calculations. Users of this method learn to be alert for such errors and to estimate their effect” [85].

In 1977, George E. Duval at WSU and Robert A. Graham at Sandia Laboratories, studied shock-induced phase transitions [86]. They annotated: “The RH equations apply even if connected states are not uniform, provided the shock front is a discontinuity in  $P$ ,  $V$ ,  $U_p$ , and  $E$  (Courant & Friedrichs [23] 1948). As a practical matter, the jump conditions are assumed to apply if field variables are changing much more rapidly in the shock front than before or behind it. Some caution must be exercised in defining shock propagation velocity if the shock front is not steady (Barker [87] 1975)”.

In 1979, Robert A. Graham critically looked at the characteristics of detectors suitable for measurements of wave profiles in shock-loaded solids [88]. In order to specify material response, he annotated: “The analysis of wave profiles requires higher-order information than the analysis of shock waves... The conservation relations for shock waves {see Eqs. (1) to (3) above} require information on particle velocity and the measured displacement-time data must be differentiated before the data specify material response. Likewise, the conservation relations for continuous wave profiles require information on acceleration, velocity gradient, or stress gradient, and time-resolved displacement or velocity data must be doubly or singly differentiated, respectively, to obtain data to specify material response”. It is a difficult task for the experimenter to provide such data without disturbing the propagation of the shock wave in the sample by the presence of several embedded detectors.

In 1980, Duane C. Wallace, a Los Alamos shock physicist, studied the compression of solids by weak shock waves [89]. He concluded that “the RH jump conditions for the increases across a shock of the normal stress, normal strain, and internal energy are not valid for weak shocks in solids. Correct jump equations for a solid can be obtained by integrating the equations for conservation of mass, momentum, and energy along the Rayleigh line through the shock process; these jump equations then depend on the details of the shock profile”. He computed a “weak-shock equation of state” and found that the pressure on the principal adiabat differs from the results of the RH theory by several percent in the range 0–100 kbar.

In 1987, Roger Blandford at Caltec (Pasadena, CA) and David Eichler at the University of Maryland (College Park, MD), two US astrophysicists, discussed the origin of cosmic rays by astrophysical shocks [90]. They annotated: “A shock front is a surface of discontinuity across which there is a steady flow of mass, momentum and energy. Of course, no shock wave is truly steady or discontinuous and all that we require is that the distance over which the flow variables vary, the shock thickness, be much smaller than the corresponding scales ahead of and behind the shock and that the overall flow pattern does not change substantially in the time it takes a fluid element to cross the shock...”

In 1993, Thomas J. Ahrens, a geophysics professor and shock physicist at Caltec, commented on the generation of shock wave data obtained for minerals [91]. He wrote: “The key assumption underpinning the validity of equations (1) to (3) {the three RH relations} is that the shock wave is steady, so that the *rise-time*  $\tau_s$  is short compared to the *characteristic time*<sup>16</sup> for which the high pressure, density, etc. are constant...”

<sup>16</sup> In compressible gas dynamics, three important *characteristic lengths* exist: (i) the *mean free path* (the average distance that a particle moves, owing to thermal agitation, between collisions) which specifies the transport properties of the gas and, compared to the size of the object or container, the flow regime; (ii) the *shock thickness* which can be very small approaching at best only a few mean free paths; and (iii) the *system length* of a shock generator which depends on its physical dimensions. An appropriate application of the RH relations requires that the system length is very large compared to the shock thickness and that quasi-steady shock parameter values exist before and after the shock discontinuity. Long

In 1997, Japanese fluid dynamicists at Kobe University and Osaka University published an extension of the RH equations for *unsteady* wave fronts of *infinitesimal rise-times* for use in condensed-matter shock physics [92]. In the following year, they also published generalized RH equations for unsteady wave fronts of *finite rise-times* which involve the combined effect of the strain waveform, its change with time, and the path in time [93]. Their analysis based on experimental data showed that the influence of the rise-time at the impacted surface depends on the texture of the studied material. In the case of a lithium fluoride (LiF) crystal this influence is negligibly small, but large for sandstone (a sedimentary rock consisting primarily of grains of quartz and feldspar, held together by a cementing matrix). Later, they also developed a theory of unsteady-state RH jump conditions using an undetermined system of partial differential equations [94,95]; derived equations of the RH type for considering geometrical effects on jumps in radial particle velocity and radial stress across outgoing spherical shock fronts; and developed a method to judge the applicability of the RH relations [96].

In 2005, William W. Anderson, a LANL shock physicist, examined the forms of the conservation relations for *unsteady* wave profiles; i.e., waves with evolving profiles. He considered situations in which, with certain assumptions, the analysis of unsteady waves in typical shock compression experiments can be taken beyond the use of the RH equations [97]. His analytical approach requires the knowledge of the wave profiles at two Lagrangian positions. He wrote: “The RH equations as commonly expressed are valid only for steady waves and their use introduces errors into the analysis of nonsteady waves... I examine the forms of the conservation relations for nonsteady wave profiles and consider situations in which, with certain assumptions, the analysis of nonsteady waves in typical shock compression experiments can be taken beyond the use of the RH equations”. He also presented a simple formula to estimate the fractional errors caused by density, stress and energy jumps due to an unsteady wave.

In 2013, André Balogh at Imperial College, (London, UK) and Rudolf A. Treumann at Ludwig-Maximilians-Universität (Munich, Germany), two space physics professors, discussed the applicability of the magnetohydrodynamic (MDH) RH relations, but some of their general comments also apply to the classical RH relations [98]. They wrote: “The RH relations give a first idea of the conditions at the shock transition. They result from the conservative character of the MHD equations which, in fact, are conservation laws for the mass flow, momentum flow and energy density... The restriction on them is that the processes which determine the generation of entropy and causing the irreversibility of the shock must all be strictly confined solely to the shock transition region. This is not so easy to achieve as it might seem at first glance. The shock transition might be much broader and more expanded than the proper shock ramp suggests. Hence the parameter values that enter the RH relations are taken correctly only sufficiently far away to both sides of the shock ramp in order to assure of not mixing-in processes that are dissipative and thus are not contained in the MHD RH conservation laws. The problem is then that the shock surface itself must be extended enough, compared with the distance from the shock where the parameter values are taken. In addition, its curvature should still be negligible in order to not destroy the assumption of shock planarity that is involved into the RH relations”.

---

*characteristic times* (such as pressure pulses above ambient) go in hand with long system lengths. For example, they can be generated in long shock tubes and blast waves produced by large explosive charges. In plane-plate impact experiments, large impactors produce long contact times between sample and impactor, hence long shock pulses in the sample.

## 4 Modifications of the classical RH relations in science and engineering

### 4.1 RH jump conditions for 1D steady shocks

The classical one-dimensional RH relations, originally developed for use in gas dynamics and supersonic aerodynamics, were extended also to other branches of physics, for example

- (i) variations of gamma ( $\gamma$ ) of a real gas with increasing energy (temperature), so-called *variable gamma RH relations* [22];
- (ii) treatment of a shock propagating into a medium with a variable local density  $\rho_0 = \rho_0(x)$  but with constant pressure  $P_0$  and particle velocity  $u_0$  [99];
- (iii) description of the downstream state of a plasma in terms of the upstream parameters; so-called *MHD RH jump conditions*, a magnetohydrodynamic analog of the classical RH relations. They obey not only the conservation equations of mass, momentum and energy, but also Maxwell's equations [100] and distinguishes between a contact discontinuity, a tangential discontinuity and a rotational discontinuity [101];
- (iv) consideration of relativistic velocities, known as the *relativistic RH relations* [102, 103];
- (v) shock solutions for non-conservation quasi-linear hyperbolic systems [104];
- (vi) derivation of the RH relations for  $\delta$ -shocks<sup>17</sup> [105, 106];
- (vii) derivation of *mean RH relations* for the change of turbulent fluxes across a shock [107];
- (viii) modeling of the detonation wave, a reactive shock wave. The *Chapman-Jouquet (CJ) theory*, a 2-zone discontinuity hydrodynamic-chemical theory, was later improved into a 3-zone theory, based upon the *Zel'dovich-von Neumann-Döring (ZND) theory* which became one of the cornerstones of modern detonation physics [108]<sup>18</sup>. Furthermore, even a 4-zone *non-equilibrium ZND theory* was developed in order to explain the various non-equilibrium processes that precede and follow chemical energy release in self-sustaining detonation waves [109]; and
- (ix) development of a *generalized version of the RH relations* for computing shocks in stellar atmospheres and winds, thereby also considering ionization, dissociation, radiation and related phenomena which are important features of very strong shocks [110].

### 4.2 RH jump conditions for 2D steady shocks

The RH relations can also be applied on 2D steady shock phenomena such as curved and reflected plane shocks. A fluid dynamical analysis, carried out by Hilary Ockendon

<sup>17</sup> Delta ( $\delta$ )-shocks which can be considered as a limit of shock waves carrying mass, momentum and energy are related not only with transport but also with concentration processes. The state variables of  $\delta$ -shocks contain Dirac delta functions which are useful for modeling shock phenomena in granular hydrodynamics such as shock propagation in dusty gases.

<sup>18</sup> In the case of a *chemical detonation*, the law of conservation of energy requires that contribution of heat by chemical reaction,  $\Delta Q$ , must be considered. An extension of the third RH equation, cf. equations (3a) and (3b), gives

$$\Delta e = e_1 - e_0 = \Delta Q + 1/2(P_0 + P_1)(1/\rho_0 - 1/\rho_1).$$

However, the first two RH relations, cf. equations (1a, b) and (2a, b), are not modified. Note that the entropy exhibits a sharp increase at the front of the detonation wave and an additional increase throughout the reaction zone.

and John R. Ockendon, two British fluid dynamicists and applied mathematicians at the Oxford Centre for Industrial and Applied Mathematics (OCIAM), revealed that the tangential velocity is conserved across a shock, while  $\rho$ ,  $P$  and the normal velocity  $u$  satisfy the RH relations which are identical for a steady 1D shock. For example, the 2D flow through a shock can be viewed as equivalent to a 1D shock with a superimposed tangential velocity [111]. Chinese researchers studied shock propagation in a current-carrying cylindrical non-neutral plasma under the influence of electric and magnetic fields and developed 2D RH relations which revealed new insight into shock upstream and downstream profiles [112].

For a system of equations in multi-dimensions, the RH relation is simply the collection of the RH relation obtained from each individual equation [113].

### 4.3 RH jump conditions for unsteady shocks

The RH relations, originally developed for a shock with a single steady wave front with an infinitesimal thickness, have also been further developed in the late 1960s by US researchers for unsteady shock waves in order to better adapt their data analysis to measured shock compression in solid targets [76–82]. In 1987, Kenneth C. Hall, an aeronautical engineer at MIT (Cambridge, MA), published unsteady RH relations for use in turbomachinery which give the nonlinear relationship on either side of the shock [114]. He derived the shock jump conditions by using Lagrangian coordinates and integrating the Euler equations from just upstream to just downstream of the shock. Because the time derivative is bounded in the shock frame of reference and the interval of the integration goes to zero, the time derivative does not appear in the shock-jump conditions. From 1997, Japanese researchers also generalized the RH relations for unsteady plane shock wave fronts of infinitesimal and finite rise-times [92–96].

### 4.4 Some examples of applications

The RH relations are widely used in S&E, for example in condensed-matter shock physics, collisionless shock wave physics, detonation physics, high-speed combustion, geophysics, sonic booms, supersonic aerodynamics, hydrodynamics, materials dynamics, traffic flow analysis, magnetohydrodynamics, rarefied gas dynamics and cosmical gas dynamics. They have been applied by experimenters to determine shock parameters of all states of aggregation and dimensions, ranging from laboratory to cosmic dimensions. Examples of studies encompass in

- (i) *gas dynamics*: motion of a shock wave through a small area change in a channel using the Chisnell-Whitham rule [115–117]; interaction of a shock wave with an upstream turbulence [118]; breakdown phenomena in gases occurring under the action of focused high-energy laser pulses [119]; condensed multi-phase flows [120] and reactive flows [121]; shocks propagating in a heterogeneous mixtures of compressible fluids [122];
- (ii) *hydrodynamics*: hydrodynamic jumps in shallow water [123]; high-speed water-solid impact [124]; derivation of the conservation equations for an incident and reflected underwater shock wave in order to determine shock-loading of submerged rigid structures [125];
- (iii) *condensed matter shock physics*: shock-compression of liquids (including simple molecular fluids obeying the Lennard-Jones-Devonshire (LJD) equation of state [126, 127] and *waterlike substances*<sup>19</sup> [128]), polycrystalline solids and single

<sup>19</sup> A *waterlike substance* is defined as one for which the internal energy  $e$  can be separated into one depending only on the density and a second part depending only on the entropy

- crystals (including shock-induced phase transitions [86,129]); behavior of porous solids [130,131]; elastic precursor decay in shock-impacted metal specimens [132]; preshock and plastic shock phenomena of shock-compressed solids [133];
- (iv) *magnetohydrodynamic*: shocks in space [134]; calculation of collisionless shocks, thereby considering ion and electron temperatures separately [135];
  - (v) *astrophysics*: studies under what solar wind conditions the RH solutions agree with observations of terrestrial bow shocks [136]; plasma shocks, particularly the RH jump conditions at the heliospheric termination shock [137]; accelerations of cosmic rays in supernova remnants using the RH relations to estimate shock speeds [138]; cosmic shock interaction phenomena, such as an interplanetary shock wave with the Earth's bow shock, a highly nonlinear process which requires to satisfy a number of MHD RH jump conditions [139]; radial outflows from astrophysical explosions [140];
  - (vi) *detonation physics*: classification of various propagation modes of detonation and deflagration in the relativistic regime [141]; and
  - (vii) *computational fluid dynamics*: The RH relations are discontinuous solutions of the hyperbolic conservation laws. A new numerical approach for a flux solver was introduced in 1998, known as *RH Riemann solver*, which takes into account source terms, viscous terms, and multidimensional effects [142]<sup>20</sup>. For example, a simple and accurate numerical solver for hyperbolic partial differential equations, based on the discretization of the RH jump condition, has been proposed and used to solve various bench-mark test cases of gas dynamics, shallow water flows, and ideal MHD [143].

## 5 RH relations vs. reality

Almost any textbook on shock waves treats introductorily the RH relations, but their numerous limitations are rarely discussed, thus the novice entering shock physics gets an incomplete picture of the applicability of these relations. Besides the general conditions made in the standard RH concept (such as conservation of mass, momentum and energy across a steadily propagating plane-parallel shock discontinuity), there exist further specific conditions which are important for the experimentalist, in the following discussed in topics #1 to #20. They depend not only on the state of aggregation and material property of the test sample, but also on the applied shock compression technique and the high-speed diagnostics. Deviations from the RH concept which can produce systematic errors  $\eta_1, \eta_2 \dots \eta_{20}$  are obviously more severe for condensed matter than for gases. In particular, the knowledge of such individual errors and their contributions to the total error is important (i) in the planning of accurate experimental techniques for providing reproducible EOS data, (ii) in a more detailed data analysis of shock-compressed materials [144]<sup>21</sup>, and (iii) in theoretical EOS modeling using experimental data.

---

$s$ , hence  $e(\rho, s) = e_I(\rho) + e_{II}(s)$ . Many substances, including water, behave in this fashion within a large range of pressures.

<sup>20</sup> The basic idea is to distribute the source terms, which also contain the viscous terms and multidimensional effects, from the cells to the cell interfaces. Then the fluxes on both sides of a cell interface are determined by the RH condition and a linearized Riemann solver.

<sup>21</sup> Precise Hugoniot data of real materials are essential for comparison purposes with other materials and for detecting the presence of a bend (in the limiting case, a kink) on the Hugoniot curve such as near a first-order phase transition or at the transition from an elastic state to a plastic state.

## 5.1 Properties of the sample material

- #1: *The medium is homogeneous and isotropic.* It is generally assumed that the shock propagates into a quiescent medium of known thermodynamic state. According to Roy F. Chisnell, a British mathematician and theoretical fluid dynamist at the University of Manchester, a non-uniform medium can be regarded as a succession of small-density discontinuities separated by uniform regions [145]. Assuming an ideal gas, he used the RH relations to calculate reflected shocks due to the passage of shocks through a non-uniform medium. Many materials such as multi-grain metals are assumed to be homogeneous if the grains are small compared to the measuring instruments active area.
- #2: *The medium is chemically nonreactive.* Extension of the conservation equations of energy to a reactive medium (such as a high explosive) was apparently first considered in 1907 by the French physicist Louis Crussard [146].
- #3: *In the medium there occur no irreversible changes during shock compression.* Thermodynamic responses of compressed solids are complicated by irreversible processes like plastic flow and fracture [147]. Shock-induced physical changes may occur at the shock front because of the large rise of temperature which not only speeds up reactions but tends to melt solids and keep liquids in the fluid state at pressures where they would normally freeze [148]. In a shock-compressed porous metal, irreversible deformations have been observed leading to a complete extinct of voids and steepening of the Hugoniot curve [149]. In various geologic materials, shock-induced irreversible changes (known as *permanent compaction*) were observed, causing steep release adiabatic curves in the stress-volume plane and high rarefaction velocities [150]<sup>22</sup>.
- #4: *Viscosity and thermal conductivity in the medium are neglected.* (i) To a first approximation the temperature rise in a shock wave will run in advance of the pressure rise but only a distance to a few interatomic distances. Shock zones have molecular dimensions, thus the neglect of viscosity and heat conduction is justified [51, 151]. However, the coefficients of thermal conductivity and viscosity will increase with pressure and temperature, thus leading to an increase of the shock thickness for strong shocks. (ii) For condensed matter shocked material, the thermal conduction is an order of magnitude slower than a pressure wave and simulations suggest that the thermal equilibrium takes a longer distance than a few interatomic distances [152]. (iii) Energy dissipation effects in shock-compressed liquids (water) were discussed under consideration that the ratio of specific heats is a function of shock temperature. Such effects (usually due to viscosity) become important at shock pressures >10 kbar [153].
- #5: *The ratio of specific heats, given by  $\gamma = c_p/c_v$ , is constant for a perfect fluid (polytropic gas), thus independent of temperature.* In 1948, a NACA theoretical study of the behavior of a diatomic gas was compared with an imperfect one in terms of the Mach number. Diagrams showed that practically no differences exist in terms of pressure and temperatures ratios up to Mach 3.5. Deviations in terms of relative errors were not published [154]. In the 1970s, NASA scientists developed a numerical code to calculate the inviscid flow around real configurations of a hypersonic research aircraft [155]. These results show that, beginning at about Mach 3 the assumption of air being a perfect gas is no longer acceptable, and real gas effects must be taken into account.

<sup>22</sup> Beginning in the early 1960, an understanding of geologic materials to shock response was crucial to the technological development of peaceful engineering uses for nuclear explosives, such as rapid excavation, gas stimulation, and retorting of oil shales. However, the so-called “US Plowshare Program” was eventually dropped in 1975 due to political and environment problems.

- #6: *There is no attenuation in the medium during shock propagation.* Shocks in gases are dissipated by collisions between neutral molecules or atoms. Spherical shock waves in air and under water propagate outward from the explosive source and become attenuated by both dissipation and geometrical spreading; i.e., showing an unsteady propagation behavior. In order to minimize the influence of attenuation, precise measurements of the shock front velocity are also crucial for obtaining precise shock parameter values when using the RH relations, particularly in the harsh environment of a nuclear explosion [30].
- #7: *The rigidity in a solid medium is neglected.* The classical RH relations describe shocks in fluids (gases and liquids); i.e., they neglect differences in liquid and solid behavior (*liquid Hugoniot theory*) [89]. In the shock compression of solids it is assumed that the stress is sufficiently isotropic so that an ordinary fluid-type EOS is valid. Most materials are capable of withstanding stress differences of only a few kilobars. Hence, where the stress normal to the front is 100 kbar or more, this stress can be taken equal to the mean stress, or pressures, to an approximation within experimental error [156].
- #8: *Dissociation and ionization processes are not considered.* They can remove energy from the dynamic degrees of freedom and increase  $e_1$  by a large amount relative to the increase in  $P_1/\rho_1$ . Consequently, the resulting density increases are considerably greater than would be predicted by the classical theory. Ionization effects at higher temperature produce thermal radiation across the shock front, leading to heating of the gas ahead [157].
- #9: *The pressure-volume relation of the sample material is time-independent.* This assumption is not true for real materials; however, they might be small in the case of shock waves, because the transition zone is very small. Two relatively simple time-dependent mechanisms are a viscosity in a fluid exhibiting a direct proportionality between the shear stress and shear rate (“Newtonian viscosity”) and stress relaxation [158]. Relaxation effects are present in gas flows with thermonequilibrium which can occur at different time scales and with different modes of internal energy [159]. Calculated relaxation times during shock compression in liquids are extremely short and of the order of  $10^{-13}$  to  $10^{-14}$  s [10]. In some solid materials the Hugoniot elastic limit (HEL) is a function of the wave propagation distance and apparently caused by stress relaxation behind the precursor front [160].

## 5.2 Properties of the shock wave

- #10: *The standard RH concept is based upon a single shock wave propagating in the medium.* In the case of a 1D weak shock wave<sup>23</sup> propagating in a solid, the entire shock front is not a single steady wave but rather consists of the steady elastic precursor which travels faster than the steady plastic wave [152].
- #11: *The shock wave is strictly one-dimensional and plane-parallel.* (i) Forbes annotated that “most of all the stress gauges are only calibrated for 1D compression and they have extra modes of response for 2D flow. This would be crystal gauges and metal foil gauges like manganin. The crystal gauges will give out voltage pulses that likely will look fine, but if some 2D flow exists additional voltage will be generated in these gauges. The resistance pressure gauges like manganin will be stretched in the 1D plane slightly adding to the calibrated resistance change that occurs for the 1D compression direction. Evaluation of

<sup>23</sup> In condensed-matter shock physics, dynamic pressures of weak shock waves range from about 10 to 100 kbar, depending on the Hugoniot curve of the selected material.

these errors is difficult, requiring additional experiments and multiple additional gauges to measure the strain occurring in a plane perpendicular to the intended 1D compression direction” [152]. (ii) 2D and 3D waves may approach an idealized 1D wave when the width of the wave front,  $W_s$ , is much less than the radius of the curvature of the front,  $R_s$ . This condition is important in the proper measurement of shock pressures in micro shock waves, but would not directly affect application of the RH relations, because they would hold for any width, even for  $W_s \ll R_s$ . However, due to geometrical spreading, shock wave parameter values of 2D and 3D waves resulting from point-like sources decay stronger with time than plane waves, thus increasingly deviate from the ideal step function as assumed in the standard 1D RH relations. Furthermore, curved shock fronts increase the entropy at the shock transition<sup>24</sup>. Martin Sichel, a professor of aerospace engineering at the University of Michigan, annotated that “as the curvature of shock wave increases, the shock structure becomes two-dimensional, and the RH jump conditions no longer hold” [161]. Dewey took the view that “the RH relations across a curved shock are the same as those for a plane shock, even for a very small radius of curvature” [162].

- #12: *The shock propagates as a steady plane wave.* This condition, certainly one of the most severe but often ignored, is difficult to verify experimentally. A *steady wave* propagates at constant velocity and unchanged in shape for all times. Thus, all shock parameter values behind the shock front are constant; i.e., all their gradients and rates are zero ( $d/dx = d/dt = 0$ ). This type of shock wave has no dispersive tendency, so perfect balance is achieved in any part of the wave. (i) Forbes annotated that “errors in the Hugoniot for well behaved materials like metals will a few per cent or less depending on the instrumentation used to measure at least two of the five shock parameter values ( $U_s$ ,  $u$ ,  $P$ ,  $v$ ,  $e$ ). The most common measurements are  $U_s$ ,  $u$  or  $P$ . In many experiments these parameters are only measured at one plane a finite distance from the impact surface. The shock wave takes time to become steady after a 1D shock entered the initial impact surface, which will increase the errors of RH parameters determined by measurements that do not prove the wave is steady. A more ideal experiment would be to have gauges at three Lagrangian planes (i.e., mass points) to measure how steady the 1D shock actually is. There are few such experiments due to the difficulty and cost of such experiments. Therefore it is impossible to put a numerical value on these errors with gauges only at one Lagrangian plane” [152]. (ii) In 3D shock waves – such as blast waves in air and shock waves produced by underwater explosions – the pressure sharply decreases immediately behind the shock front ( $dp/dt < 0$ ,  $dp/dx < 0$ ) which contradicts the RH concept of constant fluxes of mass, momentum and energy.
- #13: *The shock is a wave of finite amplitude without dissipation.* Dissipative effects prevent infinitesimal rise-times (see #17). In 1952, Harold Grad, a US applied mathematician at the Courant Institute of Mathematical Sciences of New York University studied the properties of plane steady shock waves from a mathematical point of view [163]. He wrote: “It is a classical result that the propagation of *non-dissipative waves of finite amplitude* is no longer symmetric, compressive fronts tending to steepen and rarefaction ones to broaden. In particular, a compressive front will steepen until it attains a vertical slope, after which it becomes multiple-valued and can no longer represent a physical quantity. If dissipation is introduced into the mathematical formulation, it is observed that

<sup>24</sup> One of the main properties of a curved shock (such as a bow shock) is that each streamline passing through the shock faces differential entropy rise, where a stronger portion of the shock leads to higher entropy rise than the weaker portion.

initial discontinuities are wiped out, and all disturbances tend to broaden with time”.

- #14: *The discontinuity must be a wave of compression ( $P_1 > 0$ ).* Rarefaction shock waves ( $P_1 < 0$ ), for a long time considered by thermo-dynamicists as forbidden but first predicted by the Soviet shock physicist Yakov B. Zel’dovich [164] in 1946, have been observed later such as in iron undergoing a shock-induced phase transition [165] and in the gas Freon-13 [ $\text{CClF}_3$ ] which has a large specific heat near the critical point [166].
- #15: *The shock-compressed sample material should not be disturbed by any release waves (“Taylor waves”) or reflected shocks [167].* This condition is very important in the shock loading of liquid and solid targets. Yogendra M. Gupta, a physics professor at WSU, pointed out that “the experimental measurements are typically made in the central region of the sample before the arrival of the edge rarefaction. Hence, the macroscopic condition of uniaxial strain is satisfied exactly during this time, and the shock wave appears to be extended infinitely in the lateral direction for times of interest” [168].
- #16: *The RH relations must hold between the states immediately before and behind the shock front.* The initial state in the upstream region (“0”) is normally well-known. Unsteady relaxation phenomena which occur during passage into the downstream region (“1”) should have died out until quasi-steady conditions are eventually reached.

### 5.3 Properties of the shock front

- #17: *The shock represents a mathematical discontinuity (step wave), therefore it has a zero rise-time<sup>25</sup>.* An exact Hugoniot loading is only possible with a step wave. Therefore, a very short rise-time is decisive for loading the material up to a point as closely as possible to the “true” Hugoniot curve. This important condition is often overlooked by users of the RH relations. (i) In gas dynamics, already Rankine [14] and later Lord Rayleigh [73] and Sir Geoffrey I. Taylor [83] concluded that if viscosity and heat conduction are taken into account, the shock wave would never become an infinitely sharp discontinuity, but would have a finite thickness. For air at standard pressure and temperature, calculations carried out incorporating heat conduction showed that the shock structure is several mean free paths (which is about 70 nm). Thus, the application of the RH relations is justified. (ii) Some condensed-matter shock physicists have assumed mechanical material models to explain measured rise-times. Assuming a viscous material model one can derive a shock profile with finite rise time [169] or determine the time for significant amounts of material to undergo phase transition such as that in Armco iron to take place [170].
- #18: *Dimensions characterizing the flow should be large compared to the shock thickness.* This condition, considered by numerous researchers as the key assumption, was apparently first formulated by John von Neumann and Robert D. Richtmyer in their famous paper on the inclusion of a purely mathematical fictitious viscosity in order to calculate shocks numerically [70]. Within the shock front – actually a very thin transition layer – viscous effects are important,

---

<sup>25</sup> Various definitions of the term *rise-time* are possible. According to the standard definition of a pulse waveform, the rise-time of a shock pressure pulse could be approximated by the time it takes for the pulse to linearly rise from 10% to 90% of its final value. H. Grad also considered other useful definitions of rise-times for characterizing shock transitions, such as by linear and non-monotonic profiles. He annotated that each physical variable will have its own profile, therefore also its own thickness [163].

but outside viscous effects are small on scales larger than the mean free path. They can be ignored outside the shock front and, therefore, in the RH equations. There are two conditions: (i) the overall hydrodynamic length scale  $\mathcal{L}$  should be very large compared to the shock thickness  $\Delta x_s$  (i.e.,  $\mathcal{L} \gg \Delta x_s$ ), so that shock fronts can usually be approximated as discontinuities. (ii) The overall flow time  $\mathcal{T}$  should be very large compared to the time for the flow to cross the front, so that the front is *quasi-steady* in a frame comoving with the front (i.e.,  $\mathcal{T} \gg \Delta x/U_s$ ). In gas dynamics, these two conditions are well fulfilled in the case of shock waves produced in shock tubes and blast waves observed at large distances from the source (such as arising from chemical explosions and from interstellar explosions). However, they are more difficult to realize in condensed-matter shock experiments, because the shock speed is not only much higher (by a factor of about 10 to 20 compared to the sound velocity in gases), but also the dimensions of the applied systems are much smaller, in dynamic EOS experiments ranging from submillimeters in diameter (such as in laser-driven small metal discs, known as *flyers*) to a few centimeters in diameter (such as in explosive-driven and impact-driven devices).

#19: *The thermodynamic equilibrium is reached immediately after passing the shock front.* In most practical cases, the material at state “0” is in an equilibrium before compression. The state of thermal equilibrium is supposed to still occur within the “shock width”, a short distance over which the fluid changes from its initial to the shocked state. In 1941, Hans A. Bethe and Edward Teller, two Los Alamos theoretical physicists, first discussed deviations from thermodynamic equilibrium in the shock front [171]. The mechanism for reaching equilibrium depends on the kinetics and probability of collisions, and the relaxation time for reaching equilibrium depends on pressure and temperature [172]. For strong shocks in fluids, shock widths are of the order of 10 nm or a few molecular diameters [173]. In the former Soviet Union, Lev V. Al’tshuler and collaborators, referring to large shock compression of condensed matter (mostly metals), made the assumptions that (i) equilibrium conditions are attained in times short compared to observation times; (ii) the medium compacts immediately to a void-less state; and (iii) any medium strength is negligible; i.e., rigidity of the solid is neglected, because the solid is regarded as a fluid [174]. Contrary to collision-dominated shocks, in collisionless shocks the plasma behind the shock is not in local thermodynamic equilibrium. For example, in the case of the Earth’s bow shock, the electron temperature increases only moderately across the shock, while the ion temperature jumps up by more than one order of magnitude [135].

## 5.4 Properties of the experimental technique

#20: *The applied experimental compression technique should produce for a certain period of time a plane shock wave of constant pressure.* However, depending on the inherent properties of the selected technique numerous individual reasons can produce deviations from this ideal which may be quite difficult to detect. Here are some examples:

- (i) Plane shock waves produced in a shock tube can be disturbed by transverse waves formed during non-uniform rupture of the diaphragm which generates spikes superimposed on the propagating shock (Fig. 4a).
- (ii) In a high-velocity impact, the tilt of the impactor with respect to the target can produce considerable variations between shock arrival times when a multiple arrangement of spring-loaded shock wave detectors is used [175].

- (iii) Studies have shown that in an explosive-driven flyer plate a series of accelerations, caused by shock reverberations, are produced [176]. Hence, the driver velocity may not be constant as the driver moves through a groove in the target plate for measuring shock velocity [177] and the measured particle and shock velocities will not be compatible.
- (iv) Forbes annotated that “it is commonly assumed that the release wave is a mirror image of the compression curve in  $P_1$ - $u_1$  space that is only correct for fluids that have no phase transitions or chemical reactions in them. It can be shown for solid polycrystalline metals that this release path is not symmetrical and therefore significant errors can be in the values of inferred particle velocity when measured free surface velocities are used” [152].

## 5.5 Suggestions

Possible violations of the ideal RH relations which depend on the applied experimental technique, the selected pressure range and the chosen material are obviously very complex, and it should be very difficult to calculate, or even only to estimate with sufficient accuracy, their influence on the total systematic error. Since in most practical cases unsteady shock waves are applied, it would be useful to concentrate efforts on a study of violations caused by a triangle pressure pulse; i.e., a pulse with a final shock rise-time which is immediately followed by a decrease of shock pressure (see Sect. 7.4).

## 6 Application of the RH relations on unsteady shock waves

Parameter values of real shock waves are never true step functions as assumed in the RH relations. Shock physicists developed two different methods to also apply the RH relations to unsteady waves: the *quasi-steady approach* for unsteady shock waves which, after quickly reaching thermodynamic equilibrium, maintain for a certain time a period of constant fluxes of (mostly kinematic) shock parameter values, and the *pseudo-steady approach* for true unsteady shock waves with an immediate pressure decay which, however, in the time scale of the shock rise-time, decays only slowly.

### 6.1 Quasi-steady approach

It is difficult to experimentally produce step functions in time of the shock parameter values. Two important examples of experimental “quasi-steady” shock waves are shown in Figure 4: in gas dynamics, propagating or reflected shock waves with an almost uniform amplitude can be generated for a limited period of time in a shock tube (Fig. 4a) which come nearest to Rankine’s concept of an ideal step wave. In condensed-matter shock physics, the high-velocity planar impact (Fig. 4b) is widely used to generate, after a certain rise-time and for a limited period of time, reproducible and well controllable shock parameter values.

### 6.2 Pseudo-steady approach

The problem is illustrated here in more detail in the example of a blast wave (Fig. 5). The pressure profile can be approximated mathematically by a modified Friedlander function [178, 179]

$$P(t) = P_0 + P_{\max} e^{-t/T_+} (1 - t/T_+). \quad (9)$$

For example, for a scaled distance  $Z > 40$  [m/kg],  $T_+$  [s] can be approximated by the relation

$$T_+ = 10^{0.23 \log(Z)} \quad \text{with} \quad Z = dW^{1/3},$$

where  $d$  [m] is the distance from the center of the charge and  $W$  [kg] the mass of the explosive charge [180]. The shape of a blast wave at distance  $r$  can be characterized by its rise-time at the shock front  $\tau$  and the positive pulse duration  $T_+$ . In the case of shock waves with peak overpressures  $< 100$  Pa, it was found experimentally that  $\tau \propto r$  and  $T_+ \propto r^{-0.19}$  which gives a ratio of  $T_+/\tau \propto r^{0.81}$ ; i.e., this ratio approximately increases linearly with distance [181]. It implies that at a large distance from the source ( $T_+ \gg \tau$ ), the decaying shock pressure behind the shock front stays almost at a constant level for a sufficiently long period of time. Hence, the application of the RH relations is justified for large distances from the source.

The pressure decay immediately behind the shock front can be characterized by a *time constant of initial decay*  $\theta$  given by

$$\theta = -(dP/dt)_t = 0/P_{\max} = 0.5T_+.$$

In the lower microsecond regime, the pressure decay is still very slow – for example for  $0 < t < 0.008T_+$  the pressure decay from  $P_{\max}$  is less than 1%. An observer fixed in space watching the wave passing by in the time scale of the rise-time would not yet know that the shock parameter values will eventually drop to zero. This clever assumption was first applied in terrestrial gas dynamics for large distances from the shock source so that time and space derivations of all shock parameters can be neglected in the Euler equations. Later, this view was also adopted by astrophysicists. At short distances from the source, however, this method could produce a considerable error, because time and space derivations of shock parameter values are not considered in the RH relations.

## 7 The complex problem of error estimation and future work

In the RH concept, an ideal shock wave is a mathematical discontinuity in pressure, density, particle velocity, temperature and entropy; i.e., the thermodynamic transition from the initial to the compressed state occurs immediately in zero time, the uppermost limit of a rapid transition. However, infinitesimal rise-times are not possible in fluid dynamics. Furthermore, most conditions listed in Section 5 are only partly fulfilled in practice. Therefore, strictly speaking all experimentally obtained Hugoniot data are somewhat positioned below the “true” Hugoniot. This brings up the important question of how closely experimentally determined dynamic equations of state can approach “true” Hugoniots.

In order to keep errors as small as possible, particularly in the measurement of precise Hugoniot data of solids, researchers made every effort to improve high-speed diagnostics as well as to develop new techniques for producing well-defined plane-parallel shock pulses of short rise-times and to also extend improved EOS measurement techniques up to ultrahigh shock pressures. A first review of experimental techniques and instrumentation was given by Fowles in 1972 [182]. In 1987, Ahrens at Caltech discussed shock wave generating systems such as by flyer plate impact and explosion generation for laboratory studies in geophysics and planetary physics, thereby using the RH relations in the analysis of experimental data [183]. In 1993, Lynn M. Barker and colleagues at Sandia Laboratories reviewed experimental gas gun techniques, the performance of piezoelectric, piezoresistive, and in-situ electromagnetic particle-velocity gauges<sup>26</sup> as well as the use of optical methods [184, 185].

<sup>26</sup> Forbes annotated that in condensed-matter shock experiments “an important question is how does in-situ gauges perturb the flow and, therefore, the measurements of shock parameter values. Most experiments will have the gauges at locations that are not in line with a previous

## 7.1 Error types involved

An *experimental error* results from the observed disagreement between the individual values of repeated measurements of a shock parameter value. If an experiment has small experimental errors, it is said to have high *precision* [186]. The *RMS (Root Mean Square) overall experimental error*  $\varepsilon$ , due to measurement of  $m$  shock parameter values, consists of  $m$  individual *relative experimental errors*  $\varepsilon_1, \varepsilon_2 \dots \varepsilon_m$  and is given by

$$\varepsilon = (\varepsilon_1^2 + \varepsilon_2^2 + \dots \varepsilon_m^2)^{1/2}.$$

*Systematic errors* can be detected neither by the experimentalist nor by repeated measurements. If an experiment has small systematic errors, it is said to have high *accuracy* [186]. The *RMS overall systematic error*  $\eta$ , here caused by violations of conditions made in the RH relations, usually consists of  $n$  individual *relative systematic errors*  $\eta_1, \eta_2 \dots \eta_n$  which can be functions of time. For example, in the case of a spherical blast wave, propagating in an inhomogeneous atmosphere with Mach 3 at short range, the conditions #5, #11, #12 and #17 listed in Section 5 would violate the RH concept due to temperature dependency of  $c_p$  ( $\eta_5$ ), geometrical spreading ( $\eta_{11}$ ), unsteady wave behavior ( $\eta_{12}$ ) and finite rise-time ( $\eta_{17}$ ), thus leading to an *RMS overall systematic error* given by

$$\eta = (\eta_5^2 + \eta_{11}^2 + \eta_{12}^2 + \eta_{17}^2)^{1/2}. \quad (10)$$

The *RMS total error* is then given by

$$\Delta = (\varepsilon^2 + \eta^2)^{1/2}. \quad (11)$$

Note that an improvement of error accuracy by repeating experiments and applying statistical methods in the data analysis can become quite costly and time-consuming such as in the case of large-scale shock experiments and precise Hugoniot measurements.

## 7.2 Examples of published error estimations

The following examples of published results using the RH relations and addressing error estimations might be of interest to the reader:

In 1954, Wayland C. Griffith and Walker Bleakney, two shock physicists at Princeton University (Princeton, NJ), carried out important shock tube experiments in gases. They applied the first two RH relations in the Lagrangian frame, equations (3a) and (4a), and computed pressure ratios for air by using measured data of density obtained from optical interferometry – thereby using the Hugoniot equation, equation (1a) – and of velocity in terms of the Mach number<sup>27</sup>. Their analysis showed that “in each of nine arbitrary intervals between  $P_1/P_0 = 0.95$  and  $0.15$  the average values agree to one part in a thousand corresponding to relative accuracies of  $0.1\%$  for weak shocks and  $0.7\%$  for the stronger ones. Above this range of shock strengths,  $M_0 > 2.4$ , the temperature becomes high enough to excite appreciable molecular vibrational motion in  $O_2$  and  $N_2$  so that  $\gamma$  is no longer constant”.

---

gauge, so the perturbation of the flow is minimal. A well calibrated multi-dimensional computer hydrodynamic flow code can predict the perturbations due to a gauge and will likely be the most used way of putting a magnitude on this error in lieu of running many costly experiments. Such hydrodynamic codes can also be used when the experiments are not fully one-dimensional to give some estimates of error due to divergent or non-1D flow” [152].

<sup>27</sup>  $P_1/P_0 = 1 + 2\gamma(M^2 - 1)/(\gamma + 1)$ , where in the Lagrangian frame  $M = M_0 = (u_0/c_0)^2$  and in the Eulerian frame  $M = M_1 = U_s/c_0$ .

In 1955, Roy W. Goranson, a Los Alamos physicist and previous student of Prof. Percy W. Bridgman at Harvard University (Cambridge, MA), and collaborators reported on first Hugoniot measurements of metals from 100 to 300 kbar using the RH relations [187]. In these studies, initiated at Los Alamos early in 1945, a plane compression wave was produced by a block of a high explosive which was sufficiently large to hopefully achieve constancy of pressure behind the shock front<sup>28</sup>. Pressure was measured using a tourmaline gauge, while the velocity of the compression wave through the plate and the free surface velocity at the bottom of the plate were measured using electric contactors known as *contact pins* [32]. The standard deviation for both velocities could be improved from previously 2% to 0.5%.

In the period 1955–1957, John M. Walsh and collaborators at Los Alamos continued Goranson’s experiments in the pressure range from about 150 to 500 kbar using a high explosive plane-wave generator [42, 59]. Probable errors for their presented experimental Hugoniot for metals were from 1% to 2% in compression,  $\Delta v_1/v_0 (= 1 - v_1/v_0)$ , for a given pressure. John M. Walsh and Melvin H. Rice who also used an explosive-driven shock wave but a moving-image camera for kinematic measurements determined shock velocities in order to measure Hugoniot curves in liquids [43]. They observed that the total scatter in the shock front velocity was typically 3% to 4%. Concerning errors for Hugoniot curve data they estimated that “the random probable error per data point is less than 1% in compression for a given pressure”.

In 1960, Robert G. McQueen and Stanley P. Marsh, two Los Alamos shock physicists, resuming the Los Alamos explosive-driver-plate geometry, extended the Hugoniot loci of 19 metallic elements up to 3 Mbar by applying the RH relations and the Mie-Grüneisen theory [188]. They stated that “the Hugoniot locus thus determined is subject to errors due to uncertainties existing in the zero-pressure data and the volume dependence of the Grüneisen ratio. It is estimated that these effects cause uncertainties of about 0.2% at 1 Mbar, increasing to 0.4% at 2 Mbar, in the shock velocity at a given particle velocity”.

In 1961, Stanley F. Minshall, a Los Alamos shock physicist, pointed out that “in mapping out a Hugoniot curve from experimental data it is more important to have precise measurements of the wave velocity than of the change in stress or particle velocity. The reason is that an error in wave velocity offsets the calculated Hugoniot point in a direction which is roughly perpendicular to the true Hugoniot curve, whereas an error in stress or in particle velocity can affect the calculated Hugoniot point in a direction which is usually not far from parallel to the true Hugoniot curve. Thus, errors in stress or particle velocity produce relatively small errors in the calculated Hugoniot curve, but the same cannot be said for errors in wave velocity” [189]. In the case of unsteady shock waves, the steepening process is not yet terminated (as in the case of a true step wave). Their studies showed that for samples with considerable dissipation the measurement of  $U_s$  at the free surface – such as measured electrically by using coaxial shorting pins [190] or optically by using a rotating-mirror smear camera [191] – introduced an additional error in  $U_s$ .

In 1962, Rudi Schall and Gustav Thomer, two German research physicists and pioneers of flash radiography at the German-French Research Institute Saint-Louis (Saint-Louis, France), studied explosive-driven shock compression in low-density materials using flash radiography and densitometry of the recorded absorption in the shock front [192]. They determined shock wave velocity  $U_s$  and density  $\rho_1$  radiographically. Similarly to their previous studies [193, 194], they calculated Hugoniot curves using equation (6b). They found that “a particular local density can be measured with an error of  $\pm 1.5\%$ . This allows mean values to be determined with an accuracy

<sup>28</sup> Explosive in-contact shock systems rather create a pressure pulse of quasi-triangular shape [68], Figure 4.1.

of  $\pm 1\%$  of the initial density  $\rho_0$ ". Using measured density and shock velocity data, they also applied the first two RH relations to determine other shock parameters. However, their method was later criticized that "the gases expanding laterally give rise to a curvature of the detonation wave (known as *edge effects*) so that the RH conditions are not entirely satisfied, and the measured X-ray absorption doesn't correspond to the value along the axis, but rather represents a mean value over the radial density variation. The first error is small if the diameter of the explosive is large as compared to the reaction zone length; the second error can be eliminated by a correcting analysis using the recorded curvature which permits to determine the density with respect to the curvature of the shock" [195].

In 1968, at a symposium organized by the National Bureau of Standards, Los Alamos shock physicists discussed the problems involved in the establishment of a standard for high-pressure dynamic shock wave work which requires more than precise measurements of Hugoniot data [177]. Potential sources of systematic errors can be caused by the properties of the sample material and the applied experimental technique, such (i) as the lack of thermodynamic and hydrodynamic equilibrium in the shock front which is necessary to apply the RH relations across the shock; (ii) that the driver plate may not be moving with constant velocity at the time of impact; (iii) that rarefactions from the rear surface and from edges of the assembly may give deceptively low apparent velocities; and (iv) that the thermodynamic states of the driver and target may not be identical.

In 1971, John W. Dewey analyzed blast waves produced from two hemispherical surface bursts which resulted from the detonations of 500-US-tons charges of TNT [63]. In a plot showing peak hydrostatic overpressures vs. scaled distances obtained from a large number of bursts, he compared calculated data obtained by applying the first two RH relations, equations (3b) and (4b), with experimental data obtained from his particle trajectory analysis and found pretty good agreement. Assuming an adiabatic gas flow behind the shock front and using the RH relations to calculate the pressure behind the shock he concluded that "the technique gives the flow properties to accuracy comparable with that for other forms of measurement, namely 5% to 10%".

In 1973, shock physicists at WSU discussed the properties of *steady shock waves* which are of interest to experimentalists concerned with the study of the propagation of waves of finite amplitudes in solid materials [9]. They annotated that "in the reduction of data from experiments use is made of the RH jump relations. In order that one may apply these it is necessary to satisfy one of the following assumptions: the shock can be treated as a mathematical discontinuity propagating as a steady wave. Application of these assumptions can lead to significant errors if gradients in adjacent regions are comparable to those in the shock or if time has not been sufficient for the flow to become steady and the curvature of the RH curve is large. The resolution of these difficulties still awaits satisfactory attention". They worked out a linearized perturbation analysis of the development of arbitrary profiles into the steady wave.

In 1977, George E. Duvall discussed the great value of precise Hugoniot data to the S&E community [196]. From the RH jump relations he deduced that "the RMS errors in pressure  $P$  and compression  $\Delta v/v_0$  in terms of particle velocity  $u$  and shock velocity  $D$  are given by

$$\delta P/P = [(\delta u/u)^2 + \delta D/D^2]^{1/2} = \delta(\Delta v/v_0)/(\Delta v/v_0).$$

He estimated that "the uncertainty in  $D$  is  $\delta D/D \sim 2.5\%$ . Measurements of  $u$  are probably better than this on the average, so the uncertainty in  $P$  and  $\Delta v/v_0$  in the average published data point is probably 2.5 to 3%".

In 1979, shock physicists at Lawrence Livermore Laboratory measured Hugoniot data for liquid argon in the pressure range 400 to 910 kbar using the RH relations [197].

They estimated that “the uncertainties in the impact velocity and initial densities are of the order of 0.1%. The total uncertainties in the measured shock velocities ranged from 0.4 to 0.6%; these are the dominant source of uncertainties in the pressure, density, and specific energy, which ranged from 0.9% to 1.8%”.

In 1979, Sandia shock physicists reviewed the state of investigations concerning the response of solids to shock compression and necessary considerations of error [129]. They wrote: “Most investigators estimate the precision of their recent particle and/or shock velocity measurements to be about 1%. This corresponds to an error of about 0.5% in compression and 2% in pressure. In addition to these errors of measurement, Hugoniot data are subject to a variety of errors of interpretation {as discussed by them in Sect. 3.2.4 of their paper}”.

In 1981, Livermore shock physicists reported on measurements of Hugoniot curves for Al, Cu and Ta, carried out in the shock pressure range 0.3–4.3 Mbar. They demonstrated that with the two-stage light gas gun method [198] very accurate Hugoniot data can be achieved (<1% error in pressure) [175].

In 1983, Indian researchers, reviewing EOS theories of condensed matter, also compared errors in shock pressures for various shock compression methods [65]. Depending on the applied experimental method, measurement errors ranged from 1% (using a light-gas gun) to 20–30% (using *photonic compression* such as produced by intense short-duration laser radiation).

In 1991, Charles E. Morris, a Los Alamos shock physicist, reviewed pioneering research in condensed-matter shock physics carried out in the 1950s at Los Alamos [199]. He wrote: “The RH relations are the cornerstone of shock-wave physics. They are based on the conservation of mass, momentum and energy. This allows the absolute measurement of pressure, compression and internal energy without the use of secondary standards.  $u_s$  {shock wave velocity} and  $u_p$  {particle velocity} can be measured to 0.5%. This enables pressure and compression to be determined to 1%. The Hugoniot measured in the early 1950s differ with our current Hugoniot measurements by about 1%. This was due to a recalibration of our streak cameras with an improved calibration procedure...”

In 2005, an international team of astrophysicists and shock physicists reported that when solving the RH equations, error bars and differences on particle velocity and shock velocity are amplified in a  $p(\rho)$  diagram by a factor of  $(\rho/\rho_0 - 1)$  [200].

In 2006, John M. Dewey extended the RH equations to give eleven different physical properties in terms of the ratio of specific heats [50]. He reported that “for {aerial} shocks with a Mach number greater than 3.5 this assumption {that air can be considered as a perfect gas} leads to an error of more than 1%. For shocks with a Mach number greater than 5.5 (i.e., hydrostatic overpressure of about 34 atm) the error is more than 5%”. Dewey and associates published similar results in their *RH Spreadsheets* [41].

In 2006, at the 6th International Conference on High Energy Density Laboratory Astrophysics at Rice University (Houston, TX), an international team of shock physicists reported on experimental results on X-ray and proton radiography of shock-compressed matter. They recommended the application of experimental density data for improving the precision of dynamic EOS studies based on velocity determination in order to avoid error amplification going through RH relations [201].

In 2012, Jerry W. Forbes, a US condensed-matter shock physicist at the Energetics Technology Center (St. Charles, MD), discussed in his book *Shock Wave Compression of Condensed Matter* fundamental experimental and theoretical concepts of condensed-matter shock wave physics, thereby also addressing different error types and approaches of error analysis [202]. (i) He recommended that “a preliminary error analysis be done prior to conducting an experiment so special attention is on measuring key limiting parameters... In spite of total lack of rigor for small sets of data,

error analysis shows what are the important limiting errors in an experiment”. (ii) He also provided the essential rules and illustrated error analysis on the example of measuring the free-surface velocity in an explosive flyer-plate system using a rotating mirror streak camera, thereby showing that the performance of accurate experiments requires knowledge of the source and magnitude of the various errors.

In 2014, Forbes [152] annotated that (i) “the error in the Hugoniot data for the chosen experimental technique can be computed by taking the experimental error and systematic errors of the measuring instruments into account. However, the actual error is more complex than just doing this error procedure since a material model has to be assumed in order to obtain the actual parameter (such as particle velocity) from the experiments. In other words, one must understand how the material being studied behaves under shock compression and stress release in order to interpret your experimental results correctly. Accurate material models for solids are more complex than fluid models for shock loading and stress release. Gauges that are inside the material have a set of errors that need to be accounted for, particularly if the wave is not perfectly 1D”; and that (ii) “the error in Hugoniot points depends on the technique used to measure the shock and particle velocities. It also depends if the material is a metal, plastic, or glass. Some materials are more difficult to get good Hugoniot data than others... Much of the techniques are based on obtaining Hugoniots on metals. For example, when you go to glass which might lose its strength suddenly at pressure, a free surface measurement that works well with metal will not work for such a glass”.

### 7.3 Problem of total error estimation

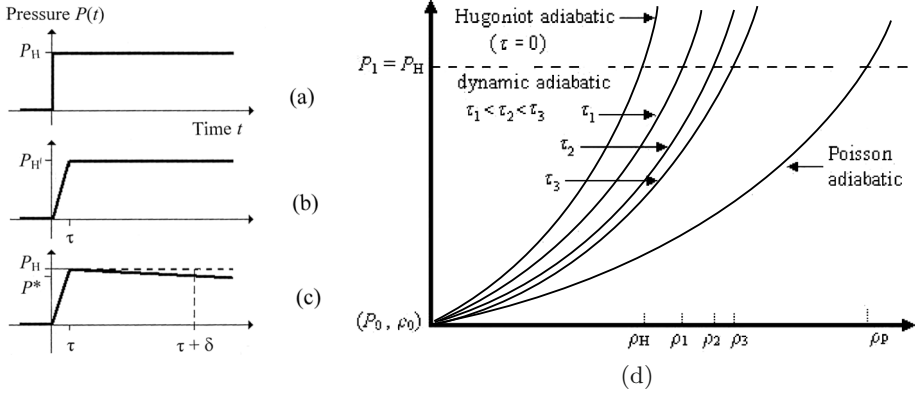
Although the RH relations are widely used in shock physics as well as in related branches of S&E, only very few users also attempted to tackle the difficult question of error estimation, a surprising result of a comprehensive literature review carried out by the author over many years. Motivations and actions were quite different. The majority of users measured two shock parameter values (in most cases kinematic parameters) and calculated the two other (thermodynamic) parameters using the first two RH relations; they either (incorrectly) considered the RH relations as being “exact” and suitable for applying on unsteady waves as well (thus neglecting all involved systematic errors,  $\eta = 0$ ) or, without proof, as being negligibly small in comparison to the allover measurement error ( $\eta \ll \varepsilon$ , thus  $\Delta \approx \varepsilon$ ).

To measure three or even all four shock parameter values with high precision and to test the admissibility of the RH relation for unsteady shock waves using the first two RH equations would require considerably more diagnostic expenditure and apparently has never been carried out. Therefore, at present practical estimations about the influence of wave unsteadiness on the total error when using the classical RH relations cannot be made.

### 7.4 Proposed numerical parametric study

For comparison purposes any numerical analysis would require a “true” shock adiabatic (or “Hugoniot”) of the sample material. Unfortunately, all known Hugoniots of real materials have been found experimentally and, therefore, are apparently not appropriate for such a crucial precision test. However, by using the theoretical Hugoniot for an ideal gas  $P_H$  ( $\rho_H$ ), equation (1a), it would be possible to make a comparison without referring to EOS data based upon experiments.

It would be of great practical interest to estimate the systematic error caused by applying the RH relations to unsteady wave behavior. Apparently, for a given material,



**Fig. 6.** Examples of dynamic pressure loading using different pressure-time profiles: (a) step wave loading (rise-time  $\tau = 0$ ) produces the shock adiabat or Hugoniot; (b) linear ramp-wave loading during time period  $\tau$  up to a constant pressure  $P_H$ ; and (c) linear ramp wave loading during different time periods  $\tau$ , followed by a linear decrease of pressure (“triangle pulse”). (d) Depending on the rise-time of ramp-wave loading, a family of dynamic adiabats in the  $(P, \rho)$ -plane is produced which are located between the *Hugoniot adiabat* and the *Poisson adiabat* (also known as *Poisson’s isentrope*). Note that with increasing  $\tau$ , ramp-wave loading up to a pressure  $P_1$  produces higher densities than shock loading. Under the assumption that at time  $t = \tau + \delta$  equilibrium is reached, pressure has decreased from  $P_H$  to  $P^*$  and the RH concept of constant fluxes is violated.

it seems to be very difficult to calculate all involved systematic errors of a particular experimental technique quantitatively and with sufficient precision. However, a reduction of the complex physical problem to a mathematical one could provide a first insight about the involved errors. What all real shock waves have in common is that they have a finite rise-time ( $\eta_{12}$ ) which in many cases is combined with an immediate rapid decay of pressure behind the shock front ( $\eta_{17}$ )<sup>29</sup>. Prominent examples are blast waves and detonation-induced underwater shock waves. In order to separately study the influence of  $\eta_{12}$  and a combination of  $\eta_{12}$  and  $\eta_{17}$  on the off-Hugoniot location of the dynamic adiabats, it might be useful to carry out a two-step parametric study:

- (i) Assumption of a linear rise of pressure loading with a rise-time  $\tau$ <sup>30</sup>, but after reaching the end of “ramp loading” at  $t = \tau$ , pressure is maintained at constant level  $p_H$  – such as produced in shock tube and plate-impact experiments for a certain time duration (Fig. 6c). For time instant  $t = \tau$ , an appropriate numerical simulation should provide a density  $\rho_{\text{ramp}}(\tau) > \rho_H$ , thus

$$\eta_{12}(\tau) = [\rho_{\text{ramp}}(\tau) - \rho_H] / \rho_H, \quad (12)$$

where  $\rho_H$  is determined from  $p_H$  using the Hugoniot equation, equation (1b). Such a study could provide a *family of dynamic adiabat curves* which, in a  $(P, \rho)$ -plot, are positioned between the *Hugoniot adiabat* and the *Poisson adiabat* (Fig. 6d). With decreasing rise-time the resulting dynamic adiabat would increasingly approach the Hugoniot curve ( $\tau = 0$ ) which requires “step wave loading” (Fig. 6a). As schematically illustrated in Figure 6d, the deviation of sample

<sup>29</sup> The subscripts refer to RH conditions #12 and #17 of Section 5.

<sup>30</sup> High-resolution measurements of shock front profiles rather reveal a  $\rho_H$ -shaped increase of shock pressure from  $P_0$  to  $P_1$ , thus the maximum pressure rate might be greater than  $P_1/\tau$  as in the case of linear pressure loading.

loading from the desired Hugoniot state is growing with increasing rise-time which – when applying the same pressure – results not only in a larger density but also in a lower sample temperature. This effect, known as *cold compression*, is increasingly used in recently developed uniaxial *quasi-isentropic compression techniques*<sup>31</sup> for “shockless” compression of a material off the Hugoniot curve by linear ramp-wave loading [203].

- (ii) Combination of a ramp-type  $P(t)$  profile rise with a  $P(t)$  profile which decays with an initial time constant  $\theta$  (Figs. 5 and 6d) – such as a nonlinear decay in the case of a blast wave, a linear decay of a N-wave sonic boom or in a sample being in-contact with an explosive (“triangle pulse”). At a time instant  $\tau + \delta$  (with  $\delta$  as a small value but chosen sufficiently long for allowing thermodynamic balance in the material), this should provide a density  $\rho_{\tau,\theta}$ , thus

$$\eta_{12,17}(\tau + \delta) = [\rho_{\tau,\theta}(\tau + \delta) - \rho_H] / \rho_H, \quad (13)$$

where  $\rho_H$  is the Hugoniot density at step wave loading up to pressure  $P_H$  using equation (1b).

Considerations on a suitable procedure for a numerical calculation of the influence of violations of the RH conditions are still in a preliminary state. Perhaps algorithms worked out in the simulation and analyses of shockless ramp-loading compression experiments may be useful for this purpose.

## 8 Conclusions

The study of a large number of journal articles, textbooks, and technical reports published within the last seventy years has revealed the following results:

- (i) The RH relations, both in their classical form and modified for special applications, are increasingly used also beyond the shock physics community, making them almost to “a maid of all work” for quickly determining shock parameter values. Their wide applicability, ranging from microscopic to even cosmic dimensions, certainly contributed much to the rapid development of shock wave physics into an interdisciplinary field by its own.
- (ii) The enormous popularity of the RH relations, their robustness and good nature are due to their first two equations which relate kinematic to thermodynamic parameter values across the shock transition. Since kinematic properties can be measured with a higher precision than thermodynamic ones, they are favorably used to determine equations of state (EOS) in shock-compressed materials. Furthermore, they do not require any knowledge of the structure of the narrow shock transition.
- (iii) The RH relations hold for plane-parallel steady waves with step wave loading; i.e., the shock has no rise-time and the fluxes of mass, momentum and energy behind the shock front are constant for all times. Pressures and corresponding densities are located on the Hugoniot curve,  $P_H$  ( $\rho_H$ ). Real shock waves, however,

---

<sup>31</sup> *Quasi-isentropic compression* refers to a loading by a smooth ramp or a sequence of small shocks. It produces a cooler compression at almost constant entropy than an abrupt (shock) loading up to a corresponding pressure on the Hugoniot curve. Ramp-wave loading techniques (i) permit the sampling of thermodynamic states not accessible by shock-wave or static pressure methods; (ii) ensure a solid state even at very high pressure; and (iii) provide data intermediate between shock wave and static pressure results. Fastest shockless compression results of aluminum up to 110 GPa (1.1 Mbar) using laser-ramp-loading within 10 ns were first reported by US researchers in 2007 [201].

are mostly unsteady waves with finite rise-times and quasi-steady or decreasing fluxes. Thus, the resulting  $P_1(\rho_1)$  data should be located on dynamic adiabatics off the Hugoniot curve. The influence of a given finite rise-time on this shift would be important to know for EOS precision measurements, but apparently this effect has not yet been studied quantitatively. Traditionally, experimental shock rise-times are in the low submicrosecond range, and it is generally believed that this almost stepwise compression will also produce “true” Hugoniot data. However, more recent “shockless” compression experiments have shown that such rise-times produce off-Hugoniot data.

- (iv) The RH relations have frequently also been used for unsteady shock waves, however, in most cases without any consideration about systematic errors resulting from the violation of the RH step wave concept.
- (v) Two procedures are in current use for the analysis of unsteady wave profiles: In the *quasi-steady approach* the two selected measured shock parameter values stay, after a finite rise-time, at an almost constant level for a limited period of time. These constant values are then used in the RH relations to calculate other shock parameter values. Such wave profiles exist in shock tubes and in a high-velocity plane-parallel impact. In the *pseudo-steady approach* wave profiles begin with a finite rise-time which is immediately followed by a rapid (linear or exponential) decrease of the shock parameter values. Examples of such “quasi-triangle” pulses are pressure pulses observed in spherical blast and underwater shock waves, the N-wave sonic boom, and the shock pressure in a liquid or solid sample being in-contact with a detonating high explosive.
- (vi) Although the RH relations are widely used in shock wave physics as well as in related branches of S&E, only a very few users also attempted to tackle the difficult problem of error estimation, thereby mostly focusing on measurement errors which range from <1% (in shock tubes) up to about 5% (in blast waves at large distances, depending on the Mach number). These studies, however, neglected considerations on individual systematic errors, thus leaving open estimations of total RMS errors.
- (vii) Causes of systematic errors are manifold, such as originated from physical and chemical properties of the medium at ambient pressure and during shock compression, and from the shape of the wave profile of a shock parameter value  $w(t)$  generated by the applied experimental compression technique. Apparently, characteristic parameters are the shock rise-time  $\tau$  and the rate immediately behind the shock front which is given by  $(dw/dt)_\tau$ . Individual systematic errors may be negligibly small, but can add up to a considerable overall systematic error which would also increase the total error.
- (viii) A critical study of the applicability of the RH equations on unsteady shock waves in regard to involved systematic errors is still pending. A future numerical parametric study of the influence of shock rise-time and flux decrease of real shock waves on the dynamic adiabatics, particularly their shifted locations in regard to the Hugoniot, would be very helpful to better illuminate this nontrivial problem.
- (ix) The RH relations, although derived for steady shock waves, seem to be well applicable also to unsteady shock waves at great distances such as in the cases of blast waves produced by large chemical, nuclear and stellar explosions. Since the unsteadiness of a nonlinear wave increases with increasing pressure and decreasing distance from the source, it would be of great practical interest to study systematic errors using the RH relations under such conditions and to compare the results with measurements of more than two shock parameter values. However, such investigations are still pending.

- (x) In order to approach “true” Hugoniot closest as possible, it is important to shock-compress materials with rise-times shortest as possible.
- (xi) Forbes annotated that “error analysis on small data sets like shock wave Hugoniot have to be subjective but what is traditionally done is to compute the error following the formulas and techniques used for large data sets, which is the subject of papers on *Theory of Error*. This is useful, since the shock wave community can check the estimates of error from shock wave work when researchers follow the same procedures and equations for determining errors. Just how much real error is involved in following this procedure is still unknown. It would take a large data set of shock data to discover how taking a small sample of the data would result in what magnitude of error for a Hugoniot determined from the small sample set vs. the large sample set” [152].

*Acknowledgements.* The author wishes to thank Dr. Jerry W. Forbes at the Energetics Technology Center (St. Charles, MD) for his numerous comments on problems of error estimations when using the RH relations in condensed-matter shock physics.

## References

1. Rankine-Hugoniot jump conditions. Wikipedia; [http://en.wikipedia.org/wiki/Rankine%E2%80%93Hugoniot\\_conditions](http://en.wikipedia.org/wiki/Rankine%E2%80%93Hugoniot_conditions).
2. Multivariable calculus – Rankine-Hugoniot jump conditions derivation. Mathematics Stack Exchange; <http://math.stackexchange.com/questions/865142/rankine-hugoniot-jump-condition-derivation>.
3. P.O.K. Krehl, Shock wave physics and detonation physics – a stimulus for the emergence of numerous new branches in science and engineering, Eur. Phys. J. H. **36**, 85-152 (2011).
4. A. Mazzia, Numerical Methods for the Solution of Hyperbolic Conservation Laws. Rapporto Tecnico No. 68, Dipartimento di Metodi e Modelli Matematici per le Scienze Applicate, Università de Padova, Italia, 1998.
5. W.C. Griffith, W. Bleakney, Shock waves in gases, Am. J. Phys. **22**, 597-612 (1954).
6. G. Ben-Dor, *Shock Wave Reflection Phenomena* (Springer, New York, 1992).
7. C. Chalons, P. Goatin, Transport-equilibrium schemes for computing contact discontinuities in traffic flow modeling, Commun. Math. Sci. **5**, 533-551 (2007).
8. D.F. Cioffi, C.F. McKee, E. Bertschinger, Dynamics of radiative supernova remnants, Astrophys. J. (Part 1) **334**, 252-265 (1988).
9. G.W. Swan, D.E. Duvall, C.K. Thornhill, On steady wave profiles in solids, J. Mech. Phys. Solids **21**, 215-227 (1973).
10. Y.J. Horie, Classification of steady-profile shocks in liquids, J. Appl. Phys. **45**, 759-764 (1974).
11. L. Euler, Continuation des recherches sur la théorie du mouvement des fluides, Hist. Acad. Roy. Sci. Belles Lettres (Berlin) **11**, 316-361 (1757).
12. G.G. Stokes, On a difficulty in the theory of sound, Phil. Mag. **33** [III], 349-356 (1848).
13. G.F.B. Riemann, Über die Fortpflanzung ebener Luftwellen von endlicher Schwingungsweite. Abhandl. Königl. Gesell. Wiss. Gött. **8** [Math. Physik. Kl.], 243-265 (1860).
14. W.J.M. Rankine, On the thermodynamic theory of waves of finite longitudinal disturbance [read Dec. 16, 1869], Phil. Trans. Roy. Soc. Lond. **160**, 277-286 (1870); Supplement. Ibid. pp. 287-288 (1870).
15. P.H. Hugoniot, Mémoire sur la propagation du mouvement dans les corps et plus spécialement dans les gaz parfaits. 1e Partie, J. Ecole Polytech. (Paris) **57**, 3-97 (1887); Mémoire sur la propagation du mouvement dans les corps et plus spécialement dans les gaz parfaits. 2e Partie, J. Ecole Polytech. (Paris) **58**, 1-125 (1889).

16. R.J.E. Clausius, Über verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie, *Ann. Phys.* **125** [II], 353-400 (1865).
17. P.O.K. Krehl, *History of Shock Waves, Explosions, and Impact – a Chronological and Biographical Reference* (Springer-Verlag, Berlin, 2009), pp. 340-341, 387-389, 1075-1077, 1094-1095, 1148-1149, 1175-1176.
18. R. Chéret, The life and work of Pierre-Henri Hugoniot, *Shock Waves* **2**, 1-4 (1992).
19. J.N. Johnson, R. Chéret, Shock waves in solids: an evolutionary perspective, *Shock Waves* **9**, 193-200 (1999).
20. M. Salas, The curious events leading to the theory of shock waves, *Shock Waves* **16**, 477-487 (2007); *A Shock-Fitting Primer* (CRC Press, Boca Raton, FL, 2010).
21. R.H. Cole, *Underwater Explosions* (Princeton University Press, Princeton, NJ, 1948)
22. C.E. Needham, *Blast Waves* (Springer-Verlag Berlin, Heidelberg, 2010), pp. 11-15.
23. R. Courant, K.O. Friedrichs, *Supersonic Flow and Shock Waves*. Springer-Verlag, New York (corrected 5th printing 1999), pp. 121-126, 336-338.
24. D.H. Weinberg, *Astronomy 825: Radiative Gas Dynamics*, Chap. 7: Shocks (Winter 2003). Dept. of Astronomy, Mathematical and Physical Sciences, Ohio State University; <http://www.astronomy.ohio-state.edu/~dhw/A825/notes7.pdf>.
25. A. Bressan, *Hyperbolic conservation laws – an illustrated tutorial* (Dept. of Mathematics, Penn State University, University Park, PA (2009); <http://www.math.psu.edu/bressan/PSPDF/clawtut09.pdf>).
26. J.W. Forbes, *Shock Wave Compression of Condensed Matter – a Primer* (Springer-Verlag, Berlin, 2012), pp. 179-185.
27. G.W. Sutton, A. Sherman, *Engineering Magnetohydrodynamics* (McGraw-Hill Book Co., New York, 1965).
28. J. von Neumann, The point source solution. In *Blast Wave*, edited by K. Fuchs, J.O. Hirschfelder, J.L. Magee, R. Peierls, J. von Neumann (Rept. La-2000, LASL, 1947), pp. 27-55.
29. L.M. Barker, R.E. Hollenbach, Shock-wave studies of PMMA, fused silica, and sapphire, *J. Appl. Phys.* **41**, 4208-4226 (1970).
30. C.E. Ragan III, M.G. Silbert, B.C. Diven, Shock compression of molybdenum to 2.0 TPa by means of a nuclear explosion, *J. Appl. Phys.* **48**, 2860-2870 (1977).
31. B. Hayes, Particle-velocity gauges system for nanosecond sampling rate of shock and detonation waves, *Rev. Sci. Instrum.* **52**, 594-603 (1981).
32. S. Minshall, Properties of elastic and plastic waves determined by pin contactors and crystals, *J. Appl. Phys.* **26**, 463-469 (1955).
33. S.R. Brinkley Jr., J.G. Kirkwood, Theory of the propagation of shock waves, *Phys. Rev.* **71**, 606-611 (1947).
34. D. Benson, An efficient, accurate, simple ALE method for nonlinear finite element programs, *Computat. Meth. Appl. Mech. Eng.* **72**, 305-350 (1989).
35. *Arbitrary Lagrangian Eulerian and Fluid-Structure Interaction: Numerical Simulation*, edited by M. Souli, D.J. Benson (Wiley, London, UK and Hoboken, NJ, 2010).
36. W. Bleakney, D.K. Weimer, C.H. Fletcher, The shock tube: a facility for investigations in fluid dynamics, *Rev. Sci. Instrum.* **20**, 807-815 (1949).
37. R. LeVeque, *Numerical Methods for Conservation Laws* (Birkhäuser, Berlin, 1992), Chap. 2.1: Integral and differential forms, pp. 14-16.
38. R.K. Tsou, Conservation equations, in *Dynamic Response of Materials to Intense Impulsive Loading*, edited by P.C. Chou, A.K. Hopkins. Rept. AD-768-416, Air Force Materials Laboratory, Wright Patterson Air Force Base, OH (1972), Chap. 2, pp. 7-42.
39. V.N. Kukudzhanov, *Numerical Continuum Mechanics* (De Gruyter GmbH, Berlin/Boston, 2013), pp. 9-13.
40. Ames Research Staff, *Equations, Tables, and Charts for Compressible Flow*, Rept. NACA 1135 (1953); Normal shock waves, pp. 6-8.
41. *Rankine-Hugoniot spreadsheets*. Dewey McMillin & Associates Ltd., Victoria, BC V8N 2A4, Canada; see [www.blastanalysis.com](http://www.blastanalysis.com).

42. J.M. Walsh, M.H. Rice, R.G. McQueen, F.L. Yarger, Shock-wave compressions of twenty-seven metals. Equations of state of metals, *Phys. Rev.* **108** [II], 196-216 (1957).
43. J.M. Walsh, M.H. Rice, Dynamic compression of liquids from measurements on strong shock waves, *J. Chem. Phys.* **26**, 815-823 (1957).
44. M. van Thiel, A.S. Kusubov, A.C. Mitchell, *Compendium of Shock Wave Data*. Rept. UCRL-50108, Lawrence Radiation Laboratory (LRL), Livermore, CA (1977).
45. *LASL Shock Hugoniot Data Bank*, edited by S.P. Marsh (University of California Press, Berkeley, 1980).
46. *SESAME: The Los Alamos National Laboratory Equations-of-State Database.*, edited by S.P. Lyon, J.D. Johnson. Rept. LA-UR-92-3407 (1992).
47. J.D. Johnson, G.I. Kerley, G.T. Rood, *Recent Developments in the SESAME Equation-of-State Library*, edited by B.I. Bennett. Rept. LA-7130, LASL, Los Alamos, NM (1978); *Equations of state – theoretical formalism*. Los Alamos Science No. 26, 192 (2000).
48. I.V. Lomonosov, K.V. Khishchenko, P.R. Levashov, D.V. Minakov, A.S. Zakharenkov, J.B. Aidun, *International shock-wave data base*. IPCP and JIHT, RAS and SNL, USA; see <http://www.ihed.ras.ru/elbrus12/program/restore.php?id=820>.
49. P.R. Levashov, K.V. Khishchenko, I.V. Lomonosov, V.E. Fortov, Database on shock-wave experiments and equations of state available via Internet, in *APS Topical Conference on Shock Compression of Condensed Matter – 2003*, edited by M.D. Furnish, Y.M. Gupta, J.W. Forbes. AIP Conf. Proc. **706**, 87-90 (2004).
50. J.M. Dewey, The Rankine-Hugoniot equations: their extensions and inversions related to blast waves, in *Proc. 19th International Symposium on Military Aspects of Blast and Shock (MABS)* [Calgary, Alberta, Canada; Oct. 1–6, 2006]. CD-ROM, publ. by Defence R&D, Suffield, Alberta, Canada.
51. E. Jouguet, La théorie thermodynamique de la propagation des explosions, in *Verhandlungen des 2. Int. Kongresses für Technische Mechanik*, edited by E. Meissner [Zurich, Switzerland; Sept. 12–17, 1926]. Füssli, Zurich (1927), pp. 12-22.
52. R. Becker, Stoßwelle und Detonation, *Z. Phys.* **8**, 321-362 (1922).
53. *Encyclopaedic Dictionary of Physics*, edited by J. Thewlis (Pergamon Press, Oxford, 1962), p. 718.
54. S.D. Poisson, Sur la chaleur des gaz et des vapeurs, *Ann. Chem. Phys.* **23** [II], 337-353 (1823).
55. A.F. Viñas, J.D. Scudder, Fast and optimal solution to the Rankine-Hugoniot problem. NASA Memorandum 86214 (May 1985).
56. S.K. Chakrabarti, *Theory of Transonic Astrophysical Flows* (World Scientific Publishing Co., Singapore, 1990), pp. 48-53.
57. A. Siegenthaler, J. Madhani, Outline of a theory of non-Rankine-Hugoniot shock wave in weak Mach reflection. *14th Australasian Fluid Mechanics Conference* (Adelaide Univ., Adelaide, Austr.; 10-14 Dec. 2001).
58. A.H. Bepp, Underwater explosion measurements from small charges at short ranges, *Phil. Trans. Roy. Soc. Lond.* **244**, 153-175 (1951).
59. J.M. Walsh, R.H. Christian, Equation of state of metals from shock wave measurements, *Phys. Rev.* **97**, 1544-1556 (1955).
60. M.H. Rice, R.G. McQueen, J.M. Walsh, Compression of solids by strong shock waves, in *Solid State Physics. Advances in Research and Applications*, edited by F. Seitz, D. Turnbull (Academic Press, New York and London, 1958), Vol. 6, pp. 1-63.
61. N. Curle, Rankine-Hugoniot law, in *Encyclopaedic Dictionary of Physics*, edited by J. Thewlis (Pergamon Press, Oxford, 1962), pp. 194-195.
62. *The Effects of Nuclear Weapons*, edited by S. Glasstone. Prepared by the US Dept. of Defense, published by the US Atomic Energy Commission (Feb. 1964), Chap. 3.72.
63. J.M. Dewey, The properties of a blast wave obtained from an analysis of the particle trajectories, *Proc. Roy. Soc. Lond. A* **324**, 275-299 (1971).
64. Prof. John M. Dewey, private communication on June 8, 2012.
65. B.K. Godwal, S.K. Sikka, R. Chidambaram, Equation of state theories of condensed matter up to about 10 TPa, *Phys. Rep. (Rev. Ser. Phys. Lett.)* **102**, 121-197 (1983).

66. E. Murr, K.P. Staudhammer, Shock wave fundamentals: effects on the structure and behavior of engineering materials, in *Shock Waves for Industrial Applications*, edited by E. Murr (Noyes Publs., Park Ridge, NJ, 1988), pp. 13-15.
67. M.A. Barrios, D.G. Hicks, T.R. Boehly, D.E. Fratanduono, D.D. Meyerhofer, J.H. Eggert, P.M. Celliers, G.W. Collins, High-precision measurements of the EOS of hydrocarbons at 1–10 Mbar using laser-driven shock waves. LLE (Laboratory for Laser Energetics) Rev. **121**, 6-21 (2009).
68. J.W. Forbes, *Shock Wave Compression of Condensed Matter – a Primer* (Springer-Verlag, Berlin, 2012), pp. 31-57.
69. Dr. Charles E. Needham, private communication on Dec. 17, 2013.
70. J. von Neumann, R.D. Richtmyer, A method for the numerical calculation of hydrodynamic shocks, J. Appl. Phys. **21**, 232-237 (1950).
71. H.L. Brode, Blast wave from a spherical charge, Phys. Fluids **2**, 217-229 (1959).
72. W. Band, G.E. Duvall, Physical nature of shock propagation, Am. J. Phys. **29**, 780-785 (1961).
73. Lord Rayleigh, J.W. Strutt, Aerial plane waves of finite amplitude, Proc. Roy. Soc. Lond. A **84**, 247-284 (1910).
74. W. Band, *Introduction to Mathematical Physics* (Van Nostrand Company, Inc., Princeton, NJ, 1959).
75. G.E. Duvall, Shock waves in condensed media. *Int. School of Physics Enrico Fermi* (Lago di Como, Italy; July 14-26, 1969). Proc. publ. in *Physics of High Energy Density*, edited by P. Caldirola (Academic Press, New York, 1971), Vol. 48, pp. 7-50.
76. G.R. Fowles, R.F. Williams, Plane stress wave propagation in solids, J. Appl. Phys. **41**, 360-363 (1970).
77. R. Fowles, *Determination of constitutive relations from plane wave experiments* (US Defense Technical Information Center, Fort Belvoir, VA, 1970).
78. G.R. Fowles, Conservation relations for spherical and cylindrical stress waves, J. Appl. Phys. **41**, 2740-2741 (1970).
79. M. Cowperthwaite, R.F. Williams, Determination of constitutive relationships with multiple gauges in non-divergent waves, J. Appl. Phys. **42**, 456-462 (1971).
80. L. Seaman, Lagrangian analysis for multiple stress or velocity gages in alternating waves, J. Appl. Phys. **45**, 4303-4314 (1974).
81. J.B. Aidun, Y.M. Gupta, Analysis of Lagrangian gauge measurements of simple and nonsimple plane waves, J. Appl. Phys. **69**, 6998-7014 (1991).
82. W.J. Murri, D.R. Curran, C.F. Peterson, R.C. Crewdson, *Response of Solids to Shock Waves*. Tech. Rept. No. 001-71, Poulter Laboratory of SRI, Menlo Park, CA (1971). Later published in *Advances in High Pressure Research*, edited by R.H. Wentorf Jr. (Academic Press, London and New York, 1974), Vol. 4, pp. 1-163.
83. G.I. Taylor, The conditions necessary for discontinuous motion in gases, Proc. Roy. Soc. Lond. A **84**, 371-377 (1910).
84. J.M. Kelly, P.P. Gillis, Shock thickness in viscoplastic solids, J. Appl. Mech. **37**, 163-170 (1970).
85. R.T. Walsh, Finite difference methods, in *Dynamic Response of Materials to Intense Impulsive Loading*, edited by P.C. Chou, A.K. Hopkins. Rept. AD-768-416, Air Force Materials Laboratory, Wright Patterson Air Force Base, OH (Aug. 1972), Chap. 7, pp. 363-403.
86. G.E. Duvall, R.A. Graham, Phase transitions under shock wave loading, Rev. Mod. Phys. **49**, 523-579 (1977).
87. L. Barker,  $\alpha$ -phase Hugoniot of iron, J. Appl. Phys. **46**, 2544-2546 (1975).
88. R.A. Graham, Measurement of wave profiles in shock-loaded solids, in *High-Pressure Science and Technology*, edited by K.D. Timmerhaus, M.S. Barber (Plenum Publ. Corp., New York, 1979), Vol. 2, pp. 854-869.
89. D.C. Wallace, Equation of state from weak shocks in solids, Phys. Rev. **B20**, 1495-1502 (1980).
90. R. Blandford, D. Eichler, Particle acceleration at astrophysical shocks: a theory of cosmic ray origin, Phys. Rep. **154**, 1-75 (1987).

91. T.J. Ahrens, Equation of state, in *High Pressure Shock Compression of Solids*, edited by J.R. Asay, M. Shahinpoor (Springer-Verlag, New York, 1993), pp. 75-114.
92. Y. Sano, Shock jump equations for unsteady wave fronts, *J. Appl. Phys.* **82**, 5382-5390 (1997).
93. Y. Sano, Shock jump equations for unsteady wave fronts of finite rise time, *J. Appl. Phys.* **84**, 6606-6613 (1998).
94. Y. Sano, I. Miyamoto, Generalized smooth and weak-discontinuous unsteady waves, *J. Math. Phys.* **41**, 6233-6247 (2000).
95. Y. Sano, T. Sano, Unsteady state Rankine-Hugoniot jump conditions, in *15th APS Topical Conference on Shock Compression of Condensed Matter – 2007*, edited by M. Elert, M.D. Furnish, R. Chau, N.C. Holmes, J. Nguyen, AIP Conf. Proc. **955**, 267-270 (2007).
96. Y. Sano, T. Sano, Jump across an outgoing spherical shock wave front, in *15th APS Topical Conference on Shock Compression of Condensed Matter – 2007*, edited by M. Elert, M.D. Furnish, R. Chau, N.C. Holmes, J. Nguyen, AIP Conf. Proc. **955**, 271-274 (2007).
97. W.W. Anderson, Jump conditions for nonsteady waves, in *14th APS Topical Conference on Shock Compression of Condensed Matter – 2005*, edited by M.D. Furnish, M.L. Elert, T.P. Russell, C.T. White, AIP Conf. Proc. **845**, 1303-1306 (2006).
98. A. Balogh, R.A. Treumann, *Physics of Collisionless Shocks* (Springer-Verlag, New York, 2013), p. 30.
99. P.L. Sachdev, *Shock Waves and Explosions*. Monographs and Surveys in Pure & Applied Mathematics, No. 132 (Chapman & Hall/CRC, Boca Ration, FL, 2004), pp. 38-39.
100. F. de Hoffmann, E. Teller, Magneto-hydrodynamic shocks, *Phys. Rev.* **80**, 692-703 (1950).
101. *Shocks and discontinuities (magnetohydrodynamics)*. Wikipedia;[http://en.wikipedia.org/wiki/Shocks\\_and\\_discontinuities\\_\(magnetohydrodynamics\)](http://en.wikipedia.org/wiki/Shocks_and_discontinuities_(magnetohydrodynamics)).
102. A.H. Taub, Relativistic Rankine-Hugoniot equations, *Phys. Rev.* **74** [II], 328-334 (1948).
103. S.M. Carioli, Solutions of the Rankine-Hugoniot relations in relativistic magnetohydrodynamics, *Phys. Fluids* **29**, 672-675 (1986).
104. X.-B. Lin, Generalized Rankine-Hugoniot condition and shock solutions for quasi-linear hyperbolic systems. Dept. of Mathematics, North Carolina State University, Raleigh, NC (June 2, 2000); see <http://www4.ncsu.edu/~xblin/preprint/shock.pdf>.
105. V.M. Shelkovich, *Delta-shocks in the Navier-Stokes system of granular hydrodynamics*. Poster presented at the 14th Int. Conference on Hyperbolic Problems: Their Theory, Numerics, Applications (HYP 2012) [Università di Padova, Italy; June 25-29, 2012].
106. V.M. Shelkovich, Concept of delta-shock type solutions to systems of conservation laws and the Rankine-Hugoniot conditions, *Operator Theory: Advances & Applications* **231**, 297-305 (2013).
107. S.K. Lele, Shock-jump relations in a turbulent flow, *Phys. Fluids A* **4**, 2900-2905 (1992).
108. M.A. Liberman, *Introduction to Physics and Chemistry of Combustion: Explosion, Flame, Detonation* (Springer-Verlag, Berlin, 2008).
109. C.M. Tarver, Chemical energy release in one-dimensional detonation waves in gaseous explosives, *Combust. Flame* **46**, 111-133 (1982).
110. H. Nieuwenhuijzen, C. de Jager, M. Cuntz, A. Lobel, L. Achmad, A generalized version of the Rankine-Hugoniot relations including ionization, dissociation, radiation and related phenomena, *A&A* **280**, 195-200 (1993).
111. H. Ockendon, J.R. Ockendon, *Waves and Compressible Flow* (Springer-Verlag, New York, 2004).
112. Y. He, X. Hu, Y. Hu, Z. Jiang, J. Lü, Rankine-Hugoniot relations of an axial shock in cylindrical non-neutral plasma, *Phys. Plasmas* **13**, 092116 (2006).
113. Y. Zheng, *Systems of Conservation Laws: Two-Dimensional Riemann Problems* (Birkhäuser, Boston, 2001), pp. 86-88.
114. K.C. Hall, *A linearized Euler analysis of unsteady flows in turbomachinery*. Ph.D. thesis, Dept. of Aeronautics & Astronautics, MIT, Cambridge, USA (May 1987); later partly publ. with W.S. Clark and C.B. Lorence in *J. Turbomach.* **116**, 477-488 (1994).

115. R.F. Chisnell, The normal motion of a shock wave through a non-uniform one dimensional medium, *Proc. Roy. Soc. Lond.* **A232**, 350-370 (1955).
116. R.F. Chisnell, The motion of a shock wave in a channel, with applications to cylindrical and spherical shock waves, *J. Fluid Mech.* **2**, 286-298 (1957).
117. G.B. Whitham, On the propagation of shock waves through regions of non-uniform area or flow, *J. Fluid Mech.* **4**, 337-368 (1958).
118. G.P. Zank, Y. Zhou, W.H. Matthaeus, W.K.M. Rice, The interaction of turbulence with shock waves: a basic model, *Phys. Fluids* **14**, 3766-3774 (2002).
119. Yu.P. Raizer, Heating of a gas by a powerful light pulse, *Sov. Phys. J. Exp. Theor. Phys.* **21**, 1009-1017 (1965).
120. E. Daniel, J. Massoni, Jump relations across shock waves in condensed multiphase flows: a comparison between numerical and analytical solutions. *18ème Congrès Français de Mécanique* (Grenoble, France; August 27-31, 2007).
121. W. Fickett, W.C. Davis, *Detonation* (University of California Press, Berkeley and Los Angeles, 1979), pp. 16-20, 98-102.
122. S.L. Gavriluk, R. Saurel, Rankine-Hugoniot relations for shocks in heterogeneous mixtures, *J. Fluid Mech.* **575**, 495-507 (2007).
123. O. Thual, Modeling rollers for shallow water flows, *J. Fluid Mech.* **728**, 1-4 (2013).
124. Y.C. Huang, F.G. Hammitt, T.M. Mitchell, Note on shock wave velocity in high-speed liquid-solid impact, *J. Appl. Phys.* **44**, 1868-1869 (1973).
125. R. Ghoshal, N. Mitra, Non-contact near-field underwater explosion induced shock-loading of submerged rigid structures: nonlinear compressibility effects in fluid structure interaction, *J. Appl. Phys.* **112**, 024911 (2012).
126. H.G. David, S.D. Hamann, Some properties of compressional waves in Lennard-Jones-and-Devonshire liquids, *Austral. J. Chem.* **14**, 372-386 (1961).
127. A. Satoh, Rankine-Hugoniot relations for Lennard-Jones liquid, *J. Fluid Eng.* **116**, 625-630 (1994).
128. R.J. Seeger, H. Polachek, On shock-wave phenomena: waterlike substances, *J. Appl. Phys.* **22**, 640-654 (1951).
129. L. Davison, R.A. Graham, Shock compression of solids, *Phys. Rep.* **55**, 255-379 (1979).
130. L. Davison, Shockwave structure in porous solids, *J. Appl. Phys.* **42**, 5503-5512 (1971).
131. L.G. Bolkhovitinov, Yu.B. Khvostov, The Rankine-Hugoniot relation for shock waves in very porous media, *Nature* **274**, 882-883 (1978).
132. P.P. Gillis, Elastic precursor decay in tantalum, *J. Appl. Phys.* **42**, 2145-2146 (1971).
133. G.E. Duvall, Shock waves and equations of state, in *Dynamic Response of Materials to Intense Impulsive Loading*, edited by P.C. Chou, A.K. Hopkins. Rept. AD-768-416, Air Force Materials Laboratory, Wright Patterson Air Force Base, OH (1972), Chap. 4, pp. 89-118.
134. J.K. Chao, B. Goldstein, Modification of the Rankine-Hugoniot relations for shocks in space, *J. Geophys. Res.* **77**, 5455-5466 (1972).
135. J.J. Sanderson, R.A. Uhrig Jr., Extended Rankine-Hugoniot relations for collisionless shocks, *J. Geophys. Res.: Space Phys.* **83**, 1395-1400 (1978).
136. D. Winterhalter, M.G. Kivelson, R.J. Walker, C.T. Russell, The MHD Rankine-Hugoniot jump conditions and the terrestrial bow shock: a statistical comparison, *Adv. Space Res.* **4**, 287-292 (1984).
137. E.C. Roelof, S.M. Krimigis, D.G. Mitchell, R.B. Decker, J.D. Richardson, M. Gruntsman, H. Funsten, D. McComas, Implications of generalized Rankine-Hugoniot conditions for the PUI population at the Voyager 2 termination shock, in *Proc. 9th Annual International Astrophysics Conference*, edited by J. Le Roux, G.P. Zank, A.J. Coates, V. Florinski. AIP Conf. Proc. **1302**, 133-141 (2010).
138. R.E. Lee, S.C. Chapman, R.O. Dendy, Numerical simulations of local shock reformation and ion acceleration in supernova remnants. *31st EPS Conference on Plasma Physics [London, UK; June 26-July, 2, 2004]*. ECA **28G**, Paper O-4.15 (2004).
139. G. Palocchia, A.A. Samsonov, M.B. Bavassano Cattaneo, M.F. Marcucci, H. Rème, C.M. Carr, J.B. Cao, Interplanetary shock transmitted into the Earth's magnetosheath: cluster and double star observations, *Ann. Geophys.* **28**, 1141-1156 (2010).

140. R.D. Blandford, C.F. McKee, Fluid dynamics of relativistic blast waves, *Phys. Fluids* **19**, 1130-1138 (1976).
141. Y. Gao, C.K. Law, *Rankine-Hugoniot relations in relativistic combustion waves*. [arXiv:1210.3455 \[astro-ph.CO\]](#) (2012).
142. P. Jenny, B. Müller, Rankine-Hugoniot-Riemann solver with considering source terms and multi-dimensional effects, *J. Comput. Phys.* **145**, 575-610 (1997).
143. S. Jaisankar, S.V.R. Rao, A central Rankine-Hugoniot solver for hyperbolic conservation laws, *J. Comput. Phys.* **228**, 770-798 (2009).
144. A. Konyukhov, A. Likhachev, V. Fortov, S. Anisimov, Nonlinear analysis of stability of plane shock waves in media with arbitrary thermodynamic properties, in *28th Int. Symposium on Shock Waves*, edited by K. Kontis (Springer-Verlag, Heidelberg & Berlin, 2012), Vol. 2, pp. 531-536.
145. R.F. Chisnell, The motion of a shock wave through a nonuniform one-dimensional medium, *Proc. Roy. Soc. Lond. A* **232**, 350-370 (1957).
146. L. Crussard, Ondes de choc et onde explosive, *Bull. Soc. Industrie Minérale* **6** [IV], 257-364 (1907); Propriété de l'onde explosive, *C. R. Acad. Sci. Paris* **144**, 417-420 (1907).
147. G.E. Duvall, Semiannual report, 1 February 1973 to 31 July 1973. Contract No.DAAG-46-C-0104, AMMRC, Watertown, MA (1973).
148. S.D. Hamann, Effects of intense shock waves, in *Advances in High Pressure Research*, edited by R.S. Bradley (Academic Press, London & New York, 1966), Vol. 1, pp. 85-141.
149. D.P. Dandekar, Behavior of porous tungsten under shock compression at room temperature, *J. Appl. Phys.* **48**, 2871-2879 (1977).
150. C.F. Petersen, W.J. Murri, M. Cowperthwaite, Hugoniot and release-adiabat measurements for selected geological materials, *J. Geophys. Res.* **75**, 2063-2072 (1970).
151. H. Eyring, R.E. Powell, G.H. Duffey, R.B. Parlin, The stability of detonation, *Chem. Rev.* **45**, 69-181 (1949), Appendix A.
152. Dr. Jerry W. Forbes, private communications on Nov. 22, 2014.
153. M. Müller, *Energy dissipated at the shock wave during its propagation in sea water*. Colloquium Fluid Dynamics, Institute of Fluid Dynamics, Prague (2007).
154. A.J. Eggers Jr., One-Dimensional Flows of an Imperfect Diatomic Gas. NACA Rept. 959 (1950).
155. F. Marconi, M. Salas, L. Yaeger, Development of computer code for calculating the steady super/hypersonic inviscid flow around real configurations. Vol. 1: *Computational technique*. Rept. NASA CR-2675 (1976).
156. G.R. Fowles, Shock wave compression of hardened and annealed 2024 aluminum, *J. Appl. Phys.* **32**, 1475-1487 (1961).
157. D.C. Pack, Shock wave phenomena, in *Research Frontiers in Fluid Dynamics*, edited by R.J. Seeger, G. Temple (Interscience Publications, New York, 1965), Chap. 8, pp. 212-249.
158. G.E. Duvall, Concepts of shock wave propagation, *Bull. Seismology Soc. Am.* **52**, No. 4, 869-893 (Oct. 1962).
159. T.-P. Liu, Hyperbolic conservation laws with relaxation, *Commun. Math. Phys.* **108**, 153-175 (1987).
160. J.W. Taylor, M.H. Rice, Elastic-plastic properties of iron, *J. Appl. Phys.* **34**, 364-371 (1963).
161. M. Sichel, Structure of weak non-Hugoniot shocks, *Phys. Fluids* **6**, 653-662 (1963).
162. J.M. Dewey, Spherical shock waves, in *Handbook of Shock Waves*, edited by G. Ben-Dor, O. Igra, T. Elperin (Academic Press, San Diego, 2001), Vol. 2, pp. 441-481.
163. H. Grad, The profile of a steady plane shock wave, *Commun. Pure Appl. Math.* **5**, 257-300 (1952).
164. Y.B. Zel'dovich, On the possibility of rarefaction shock waves, *Zh. Eksp. Teor. Fiz.* **16**, 363-364 (1946).
165. W.E. Drummond, Multiple shock production, *J. Appl. Phys.* **28**, 998-1001 (1957).
166. S.S. Kutateladze, Al. A. Borisov, A.A. Borisov, V.E. Nakoryakov, Experimental detection of a rarefaction shock wave near a liquid-vapor critical point, *Sov. Phys. Dokl.* **25**, 392-393 (1980).

167. G.I. Taylor, The dynamics of the combustion products behind plane and spherical detonation fronts in explosives, *Proc. Roy. Soc. Lond. A* **200**, 235-247 (1949/1950).
168. Y.M. Gupta, *Shock waves in condensed media*. In *McGraw-Hill Encyclopedia of Science & Technology*, 9th edn. (McGraw-Hill Book Co., New York, 2005), pp. 438-439.
169. W. Band, G.E. Duvall, Physical nature of shock propagation, *Am. J. Phys.* **29**, 780-785 (1961).
170. L. Barker, L. Hollenbach, Shock wave study of the alpha-epsilon phase transition in iron, *J. Appl. Phys.* **45**, 4872-4887 (1974).
171. H.A. Bethe, E. Teller, Deviations from Thermal Equilibrium in Shock Waves. BL Rept. X-117, BRL, Aberdeen Proving Ground, MD (1941).
172. G.E. Duvall, G.R. Fowles, Shock waves, in *High Pressure Physics and Chemistry*, edited by R.S. Bradley (Academic Press, New York, 1963), Vol. 2, p. 212.
173. G.A. Lyzenga, T.J. Ahrens, W.J. Nellis, A.C. Mitchell, The temperature of shock-compressed water, *J. Chem. Phys.* **76**, 6282-6286 (1982).
174. L.V. Al'tshuler, K.K. Krupnikov, B.B. Lebedev, V.I. Zhuchikin, M.I. Brazhnik, Dynamic compressibility and equation of state of iron under high pressure, *Sov. Phys.* **7**, 606-614 (1958).
175. A.C. Mitchell, W.J. Nellis, Shock compression of aluminum, copper, and tantalum, *J. Appl. Phys.* **52**, 3363-3374 (1981).
176. R.G. Shreffler, W.E. Deal, Free surface properties of explosive-driven metal plates, *J. Appl. Phys.* **24**, 44-48 (1953).
177. W.J. Carter, S.P. Marsh, J.N. Fritz, R.G. McQueen, The equation of state of selected materials for high-pressure references, in *Accurate Characterization of the High-Pressure Environment*, edited by E.C. Lloyd. NBS Special Publication No. 326, US Government Printing Office, Washington, DC (1971), pp. 147-158.
178. F.G. Friedlander, The diffraction of sound pulses. I. Diffraction by a semi-infinite plate, *Proc. Roy. Soc. Lond. A* **186**, 322-344 (1946).
179. W.E. Baker, *Explosions in Air* (University of Texas Press, Austin, 1973).
180. M. Larcher, *Pressure-time functions for the description of air blast waves*. JRC Technical Note 46829, Joint Research Centre, Ispra, Italy (2008).
181. H. Honma, I.I. Glass, C.H. Wong, O. Hoist-Jensen, D. Xu, Experimental and numerical studies of weak blast waves in air, *Shock Waves* **1**, 111-119 (1991).
182. G.R. Fowles, Experimental techniques and instrumentation, in *Dynamic response of materials to intense impulsive loading*, edited by P.C. Chou, A.K. Hopkins. Rept. AD-768-416, Air Force Materials Laboratory, Wright Patterson Air Force Base, OH (1972), Chap. 8, pp. 405-480.
183. T.J. Ahrens, Shock wave techniques for geophysics and planetary physics, *Meth. Exp. Phys.* **24**, 185-210 (1987).
184. J.W. Forbes, *Shock Wave Compression of Condensed Matter – a Primer* (Springer-Verlag, Berlin, 2012); pp. 68-79.
185. L.M. Barker, M. Shahinpoor, L.C. Chhabildas, Experimental and diagnostic techniques, in *High-pressure shock compression of solids*, edited by J.R. Asay, M. Shahinpoor (Springer-Verlag, New York, 1993), pp. 43-73.
186. Y. Beers, *Introduction to the Theory of Error* (Addison-Wesley, London, 1957), p. 4.
187. R.W. Goranson, D. Bancroft, B.L. Burton, T. Blechar, E.E. Houston, E.F. Gittings, S.A. Landeen, Dynamic determination of the compressibility of metals, *J. Appl. Phys.* **26**, 1472-1479 (1955).
188. R.G. McQueen, S.P. Marsh, Equation of state for nineteen metallic elements from shock-wave measurements to two megabars, *J. Appl. Phys.* **31**, 1253-1269 (1960).
189. F.S. Minshall, The dynamic response of iron and iron alloys to shock waves, in *Response of Metals to High-Velocity Deformation*, edited by V.F. Zackay, P.G. Shewmon (Interscience, New York, 1961), pp. 249-272.
190. R.E. Duff, E. Houston, Measurement of the Chapman-Jouguet pressure and reaction zone length in a detonating high explosive, *J. Chem. Phys.* **23**, 1268-1273 (1955).
191. N.L. Coleburn, J.W. Forbes, Irreversible transformation of hexagonal boron nitride by shock compression, *J. Chem. Phys.* **48**, 555-559 (1968).

192. R. Schall, G. Thomer, Flash Radiographic Measurement of the Shock Compressibility of Magnesium Alloy, Lucite, and Polyethylene. Rept. AFSWC-TDR-62-134, Air Force Systems Command, Kirtland Air Force Base, NM (1962).
193. R. Schall, Die Zustandsgleichung des Wassers bei hohen Drucken nach Röntgenblitzaufnahmen intensiver Stoßwellen, *Z. Angew. Phys.* **2**, 252-254 (1950).
194. R. Schall, G. Thomer, Röntgenblitzaufnahmen von Stoßwellen in festen, flüssigen und gasförmigen Medien, *Z. Angew. Phys.* **3**, 41-44 (1951).
195. F. Jamet, G. Thomer, *Flash Radiography* (Elsevier, Amsterdam, 1976), pp. 120-122.
196. G.E. Duvall, Problems in shock wave research [invited paper], in *Conference on Metallurgical Effects at High Strain Rates* [Albuquerque, NM; Feb. 5–8, 1973], edited by R.W. Rohde, B.M. Butcher, J.R. Holland, C.H. Karnes (Plenum Press, New York, 1973), pp. 1-13.
197. M. Ross, W. Nellis, A. Mitchell, Shock-wave compression of liquid argon to 910 kbar, *Chem. Phys. Lett.* **68**, 532-535 (1979).
198. A.H. Jones, W.H. Isbell, C.J. Maiden, Measurements of the very-high-pressure properties of materials using a light-gas gun, *J. Appl. Phys.* **37**, 3493-3499 (1966).
199. C.E. Morris, Shock-wave equation-of-state studies, *Shock Waves* **1**, 213-222 (1991).
200. G.V. Oriskov et al., Shock compression of liquid deuterium up to 109 GPa, *Phys. Rev. B* **71**, 092104 (2005).
201. G. Chabrier et al., Hydrogen and helium at high density and astrophysical implications, in *High Energy Density Laboratory Astrophysics*, edited by S.V. Lebedev (Springer-Verlag, Dordrecht, The Netherlands, 2007), pp. 257-261.
202. J.W. Forbes, *Shock Wave Compression of Condensed Matter – a Primer* (Springer-Verlag, Berlin, 2012), pp. 82-90.
203. R.F. Smith, J.H. Eggert, A. Jankowski, P.M. Celliers, M.J. Edwards, Y.M. Gupta, J.R. Asay, G.W. Collins, Stiff response of aluminum under shockless compression to 110 GPa, *Phys. Rev. Lett.* **98**, 065701 (2007).
204. P. Krehl, Measurement of low shock pressures with piezoresistive carbon gauges, *Rev. Sci. Instrum.* **49**, 1477-1484 (1978).